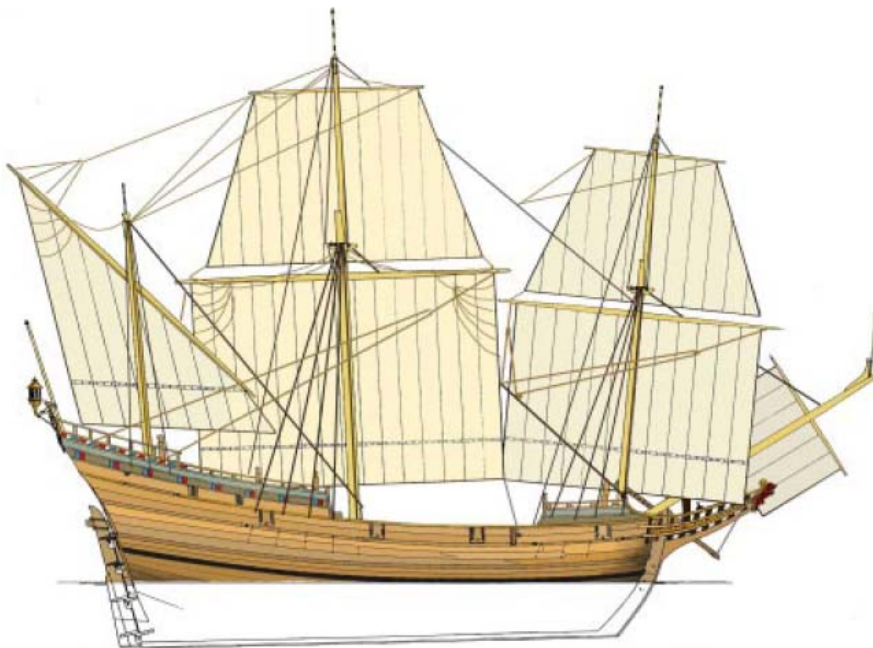


Australian National Maritime Museum

Duyfken

Volunteer Guide Notes

(Version 2)



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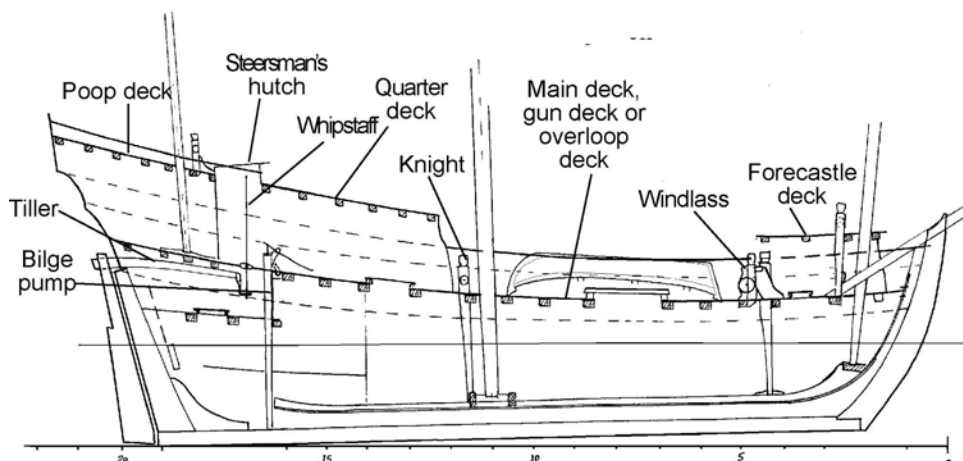
Duyfken Specifications

The Original Ship

Built	circa 1595(?), somewhere in the Netherlands
Cargo capacity	25 or 30 last (about 50 – 60 tonnes).
Crew	20
Probably armament	8–10 cannon, 3 pounders, plus swivel guns
First voyage to Indies	1601, returned Feb 1603
Second voyage to Indies	Dec 1603
Voyage to Australia	Departed Bantam, Java, Nov 1605, return reported June 1606.
Condemned beyond repair:	July 1608, at Ternate.

Specifications of Reconstructed Design

Length Keel:	16.8m
Length between stem and sternpost	19.94m
Length beakhead to taffrail	25.2m
Maximum beam	6.01m
Freeboard midships	1.8m
Height at stern, above waterline	5.5m
Draft, deep laden	2.4m
Depth in hold	2.2m
Tonnage by English rule of 1600	66.4 tons
Estimated displacement fully laden	110 tonnes.
Sail area	290 ² m (3100 ² ft)
Rig	Three-masted, square-rigged fore and main masts, lateen-rigged mizzen.
Planking below waterline	60mm oak
Above waterline	40–50mm oak
Framing timbers	400mm "room and space", 200 x 200mm floors.



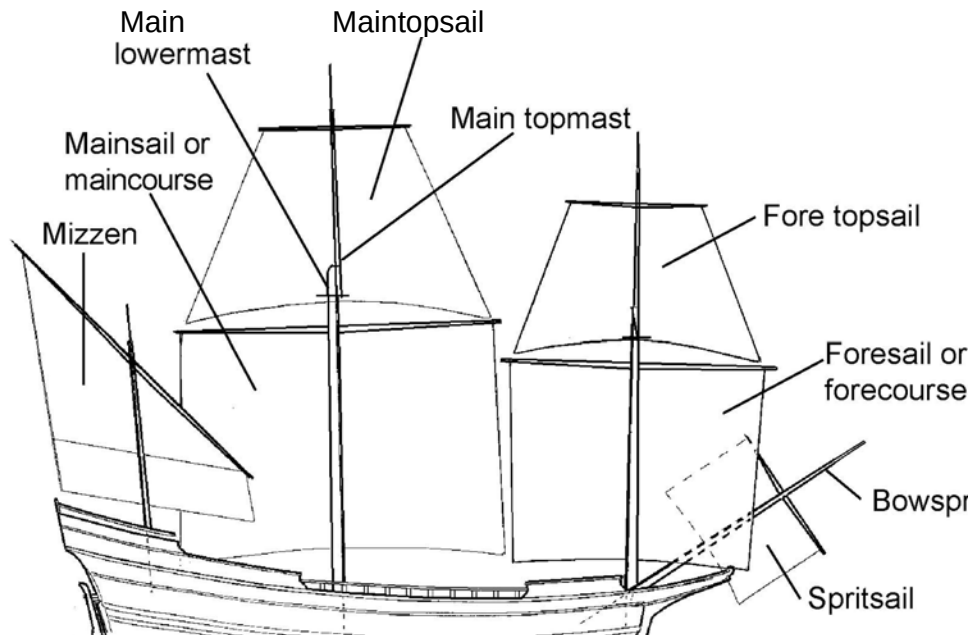
Duyfken's Rig

Like most European sea-going ships of her time, *Duyfken* had three masts and a bowsprit. She carried a small square sail under the bowsprit, but this sprit sail is nearly al-ways furled in contemporary pictures of Dutch ships. The fore and main mast each carried a "course" (lower sail) and a topsail. (Course is cognate with *corpus* meaning body — it was the main body of the rig). On the mizzen mast was a lateen sail. This sail was more to assist with steering than for propulsion. In light or choppy conditions, when sailing into the wind, the mizzen would be set to help keep the vessel's head up to the wind through "weather-vane" effect.

Later in history ships were classified by their rig (eg barque, brig, etc) but this is not appropriate to *Duyfken's* time since most of the largest ships then would have to be described as "jack-ass barques".

Duyfken's rig differed from that of the larger ships that she sailed with in that she had no "crowsnests" and her topsails and topsail yards were normally sent down to the deck when not set. Larger ships furled their top-sails aloft.

Duyfken's lower masts and topmasts on the main and foremasts (there is no mizzen topmast) might have been made from single lengths of timber, or they might have been scarfed together, but the topmasts did not stand separately, forward of the lowermasts in the way that *Endeavour's* or *Batavia's* did. *Endeavour* had "striking" or "fidded" topmasts. *Duyfken* had "pole topmasts". The use of pole topmasts, lack of "crowsnests" (properly called "tops") and the topsails sent down to the deck are all features of what was later called "polacca" or "polacre" rig. The name was applied because many of the ships carrying grain from the southern Baltic (Pomerania and Poland) were rigged in that way — polacca means from Poland, a "Polander".



Oak

Duyfken's hull is built of oak (scientific name *Quercus robur*) from Latvia in the Baltic region. The original *Duyfken* was certainly built of oak, and it is quite likely that the oak came from the Baltic region.

Oak is a hardwood and it is fairly hard. Balsa wood is a hardwood, but it is not hard. Huon pine is a softwood, but it is very hard.

Hardwood/softwood is a technical distinction between the timber from coniferous trees (softwood) and all other timbers. Conifers are the trees with their seeds in cones and with needles rather than broad leaves, all conifers except larch are “ever green”.

Most non-conifers are deciduous — they lose their leaves at a certain time of year. The cellular structure of conifers is different from that of broad leaved trees — conifers are an older type of tree.

Oak is strong, reasonably easy to work, resistant to splitting, resistant to rot. It expands and contracts with changes in moisture content, but not as much, and not as quickly as some hardwoods such as jarrah.

Building a Dutch ship plank-first

It is believed that wooden boats and ships developed from the simple dugout canoe in many parts of the world. The canoe would be extended by adding a row of planks, then a second row of planks. Over time the dugout base would decrease in size and the planked shell become the main structure of the hull. Eventually the dugout canoe is reduced to a keel. In this development, as the planked part of the hull gets bigger, strengthening structure has to be added. This can be cross beams or frames shaped like ribs. The strengthening structure is fitted into the shell of planks.

Many types of ships have been built in this way — the Classical era ships of the Greek and Romans, Viking ships, Chinese junks, Arab dhows, for example. Ships built in this way can be built by eye. The shipwrights gradually sculpt the shape of the ship.

But in Europe, in the Mediterranean region, another system developed during the Middle-Ages. In this system, the frames are erected first. Each frame has to be shaped to create the cross-section shape of ship at the point where it is erected. Some kind of plan or mathematical system is needed for working out the shapes of these frames.

The frame-first system of construction was spread from the south of Europe to northern Europe during the early 16th century. Henry VIII brought shipwrights from Portugal to show his shipwrights how to build frame first.

But the Dutch did not take to the new method. Instead, they modified their ancient plank-first, by-eye, system of construction to build current styles of large ships.

Curiously, the Dutch produced much better ships than anyone else in Europe. Their ships were economically built, sailed well, were not too deep and carried plenty of cargo. The newly formed Dutch nation, at the end of the 16th century, owned and operated much more shipping than any other European nation.

Here is a passage from a letter, probably by the merchant Pedro Lopez de Soto written in 1631 or 1632 and translated by Paulo Monteiro:

" . . . I arrive at the conclusion that the Flemish nation has understood, better than any other, maritime matters, and they practice them with much more perfection than any other (nation), and they go with their ships all over the world and to places where ours cannot navigate, because of the great storms there are in those regions, and the Turks from Argal and Zale, practicing piracy in the coasts of Italy and Spain, will not use any-thing other than Flemish [Dutch] ships . . ."

It was not until the second half of the 18th century that naval architects started to use lines like contours on a map to design the shapes of ships and the builders of frame-first ships who used plans could produce ships of shapes as successful as the Dutch who built by eye.

The *Duyfken* replica has been built as a program of experimental archaeology. She has been built in the way that the original ship would have been built — an undertaking which has been a huge leap in the dark for 20th century shipwrights who have been trained in a quite different tradition. *Duyfken* has been built plank-first, with no frames to pre-determine the shape of the hull. Oak timbers which form the outer plank-shell, some of them more than 100mm thick, have been bent to shape by heating them over open fires until the timber becomes plastic. This has been done by shipwrights working entirely by eye. Learning the technique was very difficult and some valuable oak planks were burned, but the ship-wrights have made it work. They became skilful at the ancient practice of plank-first construction, and have produced a very beautiful ship.

Building A Dutch Ship Before 1600

A talk by Mr. Bill Leonard Master Shipwright - Duyfken, on 19 February 1997 (Construction had barely started).

On seeing the title of this talk I have been asked to give, I really didn't know whether to laugh or cry. Cry - because of the lack of information available, and laugh because the entire substance of this talk may last all of two minutes. Shipbuilding before the Sixteenth Century is the question the Professors of Maritime Archaeology don't like to be asked.

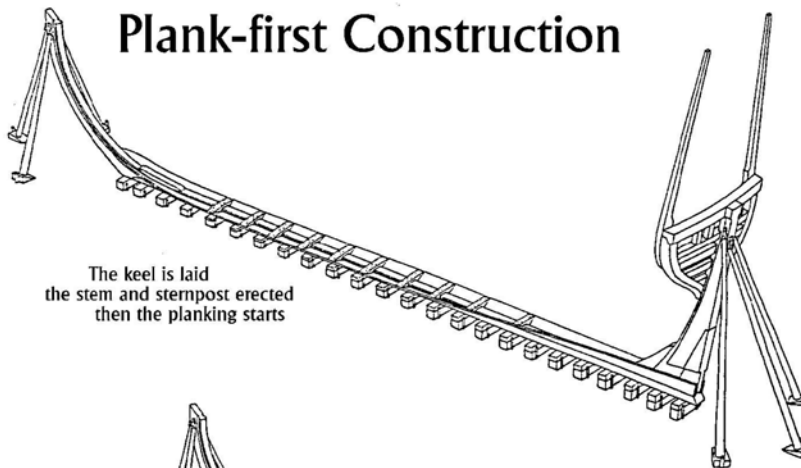
It is also the trick question that students of Maritime Archaeology see in their exam papers and tremble.

We probably know more about the construction techniques of the Roman shipwrights than those of the nuts and bolts of the everyday ship construction that took place in the 16th Century Dutch shipyard. Hopefully, of course, the *Duyfken* will answer many of questions that the sometimes abstract world of Maritime Archaeology asks.

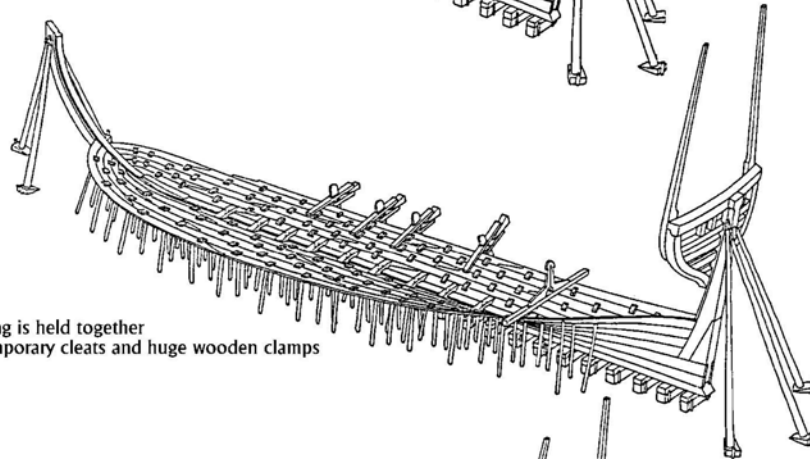
What clues do we have? A few notes, some drawings, books of formula, a few interesting contracts and, of course, those magnificent fragments of the early Dutch shipwrights — the most famous in the Southern Hemisphere being our own Batavia (the remains of the Batavia shipwreck are in the Western Australian Maritime Museum

Its always wonderful to watch people walking into the Batavia Gallery. They wander in to be confronted by a wall of wood, with interesting little openings near the top. They walk around to the foreshore or look down from the gallery. What do they see? A confusion of random shapes, a shambles of wood piled upon each other. Let's face it — it doesn't make a lot of sense. What they are viewing is the art of the ancient Shipwrights trade. The average person, looking at the Batavia now, full of disbelief, wonders why! To modern day shipwrights like myself, these magnificent relics, like the Batavia, give the answer to how. They were built by Master Shipwrights.

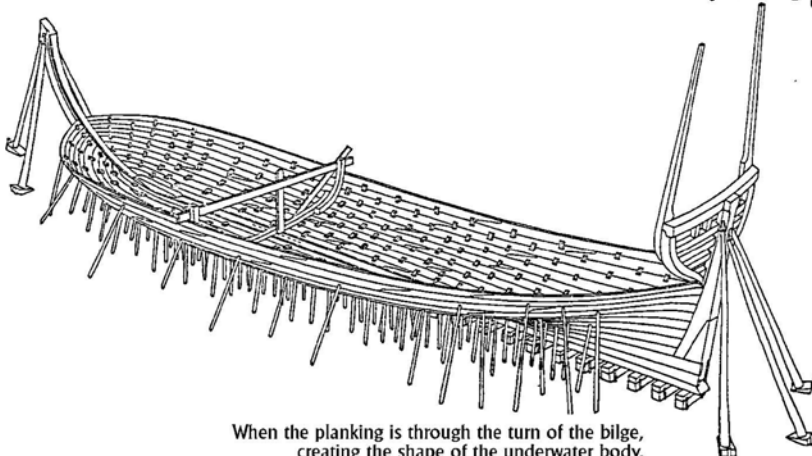
Plank-first Construction



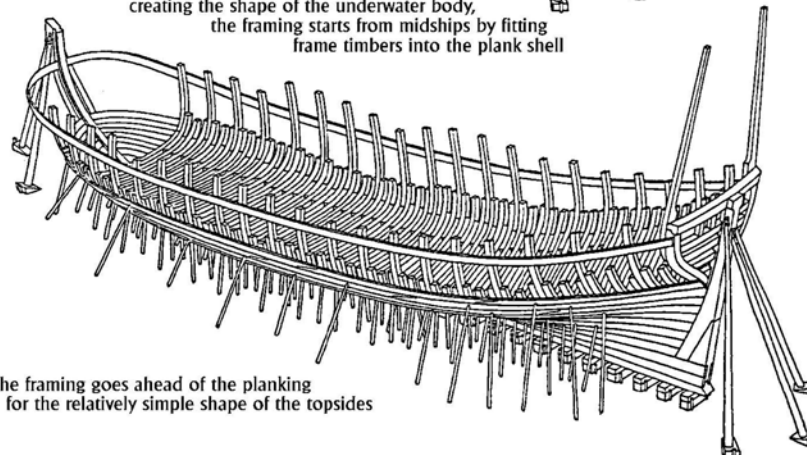
The keel is laid
the stem and sternpost erected
then the planking starts



The planking is held together
by temporary cleats and huge wooden clamps



When the planking is through the turn of the bilge,
creating the shape of the underwater body,
the framing starts from midships by fitting
frame timbers into the plank shell



Then the framing goes ahead of the planking
for the relatively simple shape of the topsides

Master Shipwrights in those days were people of substance, highly educated and organized. As an example, let's take the famous 16th Century English Shipwright Phineas Pett. Born in 1570, starting boarding school at Rochester and later, in his teens, studied at Greenwich before completing his four year degree course in 1590 at Emanuel College, Cambridge. He then was fortunate to acquire a Shipwright Apprenticeship at Deptford eventually falling in with Matthew Baker in 1595. This, of course, is the famous Matthew Baker whose ancient ships' drafts are available for all to see in a manuscript known as "A Fragment of Ancient Shipwrightery". To reiterate, Master Shipwrights of the 16th Century were bright professional people — unlike some of their modern brethren.

So how did they start to build a ship — by first of all having in their hand, the contract.

Let's firstly set the scene. A knock on the door and there before the shipwright stands a wealthy merchant of Amsterdam. "Master Shipwright I need a ship of great burden. Of course it will be grander than that old hooker, that so-and-so just returned from the East Indies in. This ship will reflect my position at Court. I intend to make my fame and fortune in the East Indies. "The shipwright reaching down for a piece of parchment, writes down the price and hands it over to the merchant. On recovering, the merchant in his best double Dutch exclaims "Sir you charge like a wounded bull!!" The shipwright, of course, acts most indignant and states "Sir I care not for fame, however I intend to make my fortune right here!!" Centuries have come and gone but some things never change.

The next question asked by the merchant is the **big** one The now sheepish merchant asks "What do I get for my money?"

Contracts are a really important clue for the building process of early ships because they often describe, in sequence, the events of the construction. (The scantlings of materials, how they were joined — fastenings to be used.) There are 4 or 5 of these 16th Century contracts available for researchers, all you need to know is how to speak 16th Century Dutch!

Personally I really wish there was more documentation available on 16th Century Ship production. I am given to believe that the Batavia took 9 months to launch — that shows very real organizational skills not a confused process of construction, not a haphazard shambles of wood, but an amazing degree of pre-planning, a complete understanding of the shipbuilding process, a well organized and controlled labour force. They knew exactly what they were doing — and knew that mistakes cost money.

With his contract in hand, the builder would have a meeting with other important people — suppliers of wood, sparmakers, blockmakers, sailmakers, riggers, blacksmiths, suppliers of tallow, tar and oils. He would get things organised and moving. Finally his thoughts would settle on the keel or probably the number of keel blocks required or possibly the condition of the stocks (the slipway foundations) that the keel blocks secure to. At this stage, surprisingly, a launching procedure is clarified in his mind.

His previous experiences and intuition coming to the fore. No point of building it if he cannot get it into the water!!!.

Soon the sawers in the pits would start sawing timber for the keel. The dimensions of which were given by the shipwright from his book of proportions and formula. Perhaps these dimensions were passed down from his father, however personal experience would have been his guide.

Passing the keel pieces over to the shipwrights, comments like “How do they do it” are heard in the yard. Perfect straight cuts reflect the sawer’s skill. The shipwright joining these pieces together with a tapering scarf joint then proceeds to cut out the rabbet, which is the recess in the keel that takes the lowest plank, the garboard strake. The bevels and angles are known to the shipwright, the final trimming of which is determined by the twist of the plank itself as it reaches the stem and sternpost. The shipwright’s concerns could revolve around the supply of good compass (naturally curved) timber. He certainly cannot build the ship without it, and having to select it himself will be a time consuming process, You see, he needs it by the cartload!! The first major piece of compass timber required is the stem, and its shape was determined by a couple of sweeps of a line, or in many cases, by the curvature of the tree itself. On larger ships, the stem would have been built up of 3 or 4 pieces.

Down at the other end of the keel, the shipwrights will have started the sternpost framing. Consisting of the sternpost itself, and compass timber in the form of fashion pieces, wing transoms, transoms, lighter pieces of wood in the form of stanchions and the long narrow top timbers that extend this prefabrication right up to the stanchions on the upper deck. This massive structure is not only a major structural element of the ship but also defines the shape of the stern, and like the stem, is secured to the keel with the help of large knees that are fastened in place with bolts. And with the stem, being set up on the keel completes the major centre line structure of the ship.

During this period, ships constructed in Holland were built shell first. That is a ship was planked out to the “runkheads” or “wrongheads” (the turn of bilge) prior to any framing timbers being installed. The garboard strake at the bow tend to be sharply twisted and bent, and would have

been formed with the help of an adze from homegrown curved timber and then steamed or at least baked over a fire. To make it bend to its final curve, the shipwrights used a *Klass*, an iron device, a sort of adjustable fork that they engaged over the edge of the plank to twist it. The timbers may have been thoroughly soaked in water, the end would have been secured to the end of a bending frame and while a fire was ignited beneath it, men using *Klasses* would force it into the designed shape.

The garboard would then be nailed to the rabbet on the keel and the stem and sternpost, while, cleat like blocks on the keel, ensure a correct angle between it and the strake. Steamed over fire, the planking continued with the next strake being fastened with temporary cleats to the garboard, props or “shores” were also employed to help create the proper shape.

As the planking extended outboard, clamps and chains were used to pull the planking together and this method of construction if applied by skilful shipwrights could produce wholly satisfactory results. The question of symmetry would often arise and this would be checked constantly by the use of a long plank and a string level extending over both sides of the ships.

Having reached out to the turn of the bilge, the shipwright would then install perhaps three sets of floors (frame timbers). I say “perhaps”, because each Master Shipwright in each yard had their own procedure, own secrets for building their vessel. Hopefully Duyfken will let us explore these construction pathways and secrets won't be secrets anymore.

From these initial floors, frames would extend up-ward to the full height of the ship's sheer. They would then be interconnected fore and aft by a series of rib-bands that extend from the fashion piece top-timbers in way of the sternpost, to the stern itself. What we have now is something that looks like a ship.

At this stage the rest of the floors are installed, as well as the remaining frames, hawespieces (framing around the bow), counter framing, 'tween chocking and port transoms. After fairing, the planking can continue apace to the top of the sheer, the only constraint being that the external planking is through fastened to the internal planking. The fastenings used are trunnels — length of dowelling with wedges drive in each end. The remaining centreline structural members can also be in-stalled, keelson (an internal keel that sits on top of the floors), apron, various breasthooks, deck hooks, transom, knees, mast straps, etc.

As the internal planking (ceiling) extends up so it meets the clamps. These are, in effect, heavier planks that support the deck beams, and

depending on the width of the ship, these beams may in fact have pillars and knees for support. Between these beams, there are also installed what are called carlines and ledges that provide support to the beams and deck above. The deck would then have been laid, possibly starting with the heavier waterways and tying strakes. On being laid, the completed structure would have been caulked with oakum, the seam being finally paid with pitch.

The ceiling would then be through fastened to the external planking as it extended up towards the upper decks and again a similar process continues with clamps, beams, carlines ledges, hanging knees, waterways, tiestrakes, caulking, pitching seams.

In the meantime other important components are being made for the ship prior to being installed. The rudder, whipstaff, windlass, capstan if she had one, beakhead, tree pumps, hatch, gratings, etc.

The internal fit out starts to loom. The owners possibly taking a special interest in the accommodation. The yard blacksmith and his team of helpers have been busy during the construction process with the supplying of fastenings, ringbolts, hinges, hoops for carriages, rings for spars, forelocks bolts, perhaps even the ship's anchors.

The sparmakers have converted the trees into spars, the sailmakers have completed the sails, the riggers having completed as much preparation as possible, wait in anticipation of a successful launch. Then they can step the spars and rig her for her working life. The hull having been caulked perhaps with calf hair and pitch. Before the launch, the underwater parts of the hull would have been perhaps dressed with a mixture of tar and white lead to prevent or lessen fouling.

The launching ways would have been greased with tallow or whale oil, the ship could have been eased over to her bilge and with the use of tackles finally launched. Looking at it, the Master Shipwright would be considering his final fit out or perhaps looking, observe that wooden sailships are the finest things from which dreams have ever been made.

Designing a Ship For Exploring Unknown Shores

How Good was Age of Discovery Ship Design? No original plans of any ship from the Age of Discovery exist because shipwrights did not use plans drawn on paper or parchment. The only plans were in the master-shipwright's head and ships were built by eye.

Replicas or reconstructions of several Age of Discovery ships have been built in recent times. They look fine and romantic, but very few of them can sail anything like as well as the original ships did. There is clearly something wrong with the way we understand the design of these ships.

From the outset, one of the stated objectives of the Duyfken Replica Project has been to produce a reconstruction that sails well enough to emulate the achievements of the original *Duyfken*.

The Research

Research to reconstruct the design of *Duyfken*, a Dutch *jacht* built at the end of the 16th-century and used for exploratory voyages, has drawn on five major types of data.

- Contemporary artwork showing sailing ships (the “iconography”).
- Ship builders’ contracts specifying the sizes and types of timbers used in ships’ construction.
- The remains of contemporary Dutch ships discovered by archaeology.
- Log books and other documents revealing the performance of the ship.
- Computer modelling of ship performance and stability.

The first objective in our research was to learn to draw ships with the right external style and appearance. We compiled an extensive catalogue of reproductions of Dutch marine art from the time of *Duyfken* and used mathematical and statistical analysis to describe and find typical forms and ratios of proportions. This goes by the rather grand name of “Morphometric Analysis of the Iconography”.

Although this analysis was aimed primarily at learning the style rather than the technical design of the ships, we began to notice that some proportions shown in the iconography consistently indicated a hull-form very different from ships of half-a-century later.

In the absence of any plans for a ship of our period there have been a number of theoretical (on paper) re-constructions designed by using the relatively narrow and box-like hull-form of the late 17th-century, whence plans are available, and combining this with the high-stern and large billowing sails of the 16th-century galleon. The results could be disastrous if tried at full-size.

Following clues in the iconography, we began tentatively to move towards a design with more beam, and with longer and sharper bow and stern; but without questioning the shallow boxy cross-sectional shape.

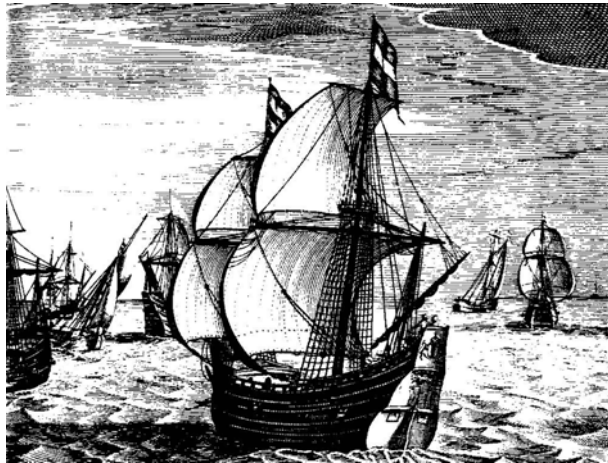
At each phase, reports on the research were sent to experts in the Netherlands for comments and criticism. A very positive response came in the form of confidential information about the on-going under-sea investigation of a *circa* 1590 shipwreck in Dutch waters which showed very strongly the long and relatively sharp bow and stern that we had tentatively reconstructed. Further-more it provided evidence for an unexpectedly sharp cross-section shape (hollow garboards). Other evidence has supported this interpretation. It is the solid evidence of

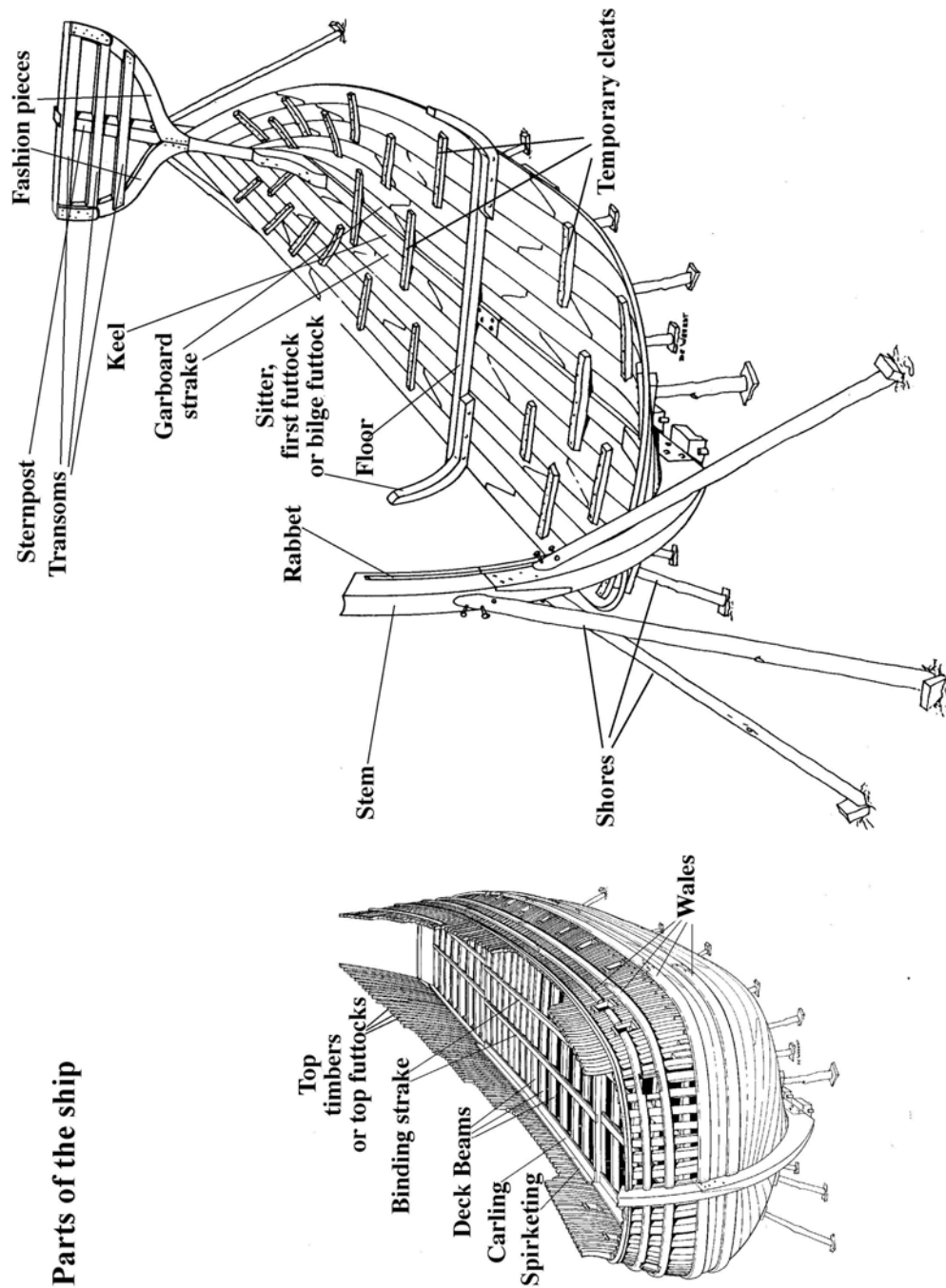
archaeology that lets us see what other evidence was hinting to us.

The design, a broad-beamed but sleek design, was tested by computer modelling (using the Western Australian developed *Maxsurf* application) and the results were most encouraging, showing a design with good stability and able to maintain good average speeds even in light conditions, in keeping with the historical evidence.

Meanwhile the original *Duyfken* kept trying to sail ahead of us—we designed a vessel that should sail well, but archival research increasingly showed that the original ship was exceptional—often out-sailing much larger ships, and manoeuvrable enough to be taken close to unexplored lee-shores even during the stormy and dangerous monsoon.

The ship as built is not identical to the design on paper and computer. She was built plank-first, using fire to bend the planks, and built partly by eye, as the original ship would have been. So the ship as built by the shipwrights should be more authentic than anything designed on paper and the natural curves of the hull will be subtly better.





Martnets

The following is a technical discussion of the ropes which help furl *Duyfken's* lower sails (the courses) when the yards are lowered. Martnets were used in the 16th century but disappeared in the early 17th century when it was no longer normal practice to lower the course yards.

Martnets are a potent source of confusion. The literature shows a number of arrangements that won't work effectively.

Anderson (1955) wrote:

They are about the most difficult of all pieces of rig-ging, not only on account of their complicated nature, but because artists have treated them so badly.

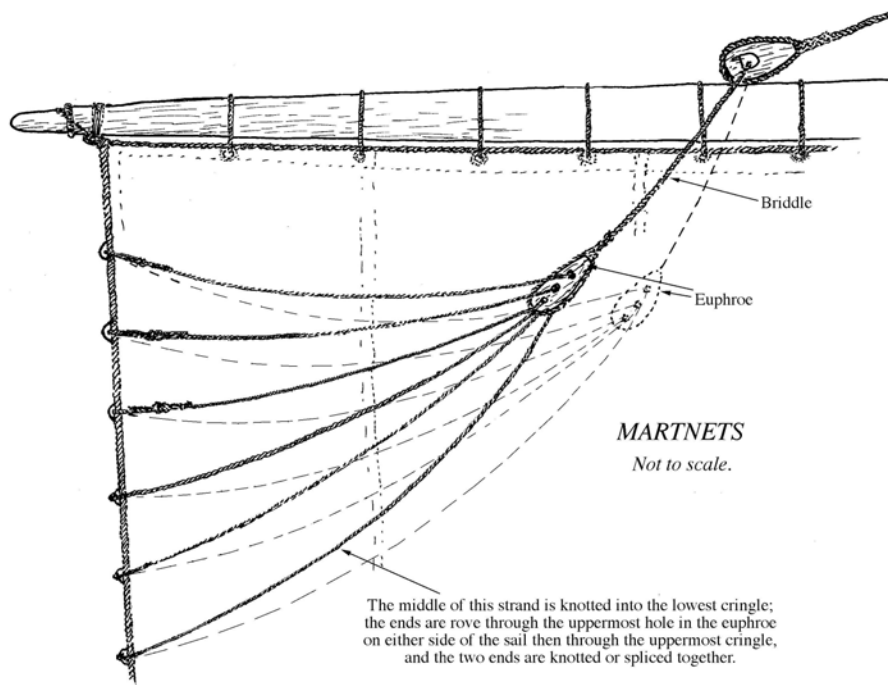
The artists who drew martnets often drew things that wouldn't work.

English martnets apparently lead through blocks at the topmast hounds which gave them plenty of room to work, but Dutch martnets always lead to blocks hung under the hounds, seemingly giving them little room to work. Anderson suggests that the martnets rove under the hounds were meant to work when the yard was lowered to furl the sail, in effect like lazy-jacks. The top-mast rove martnets were introduced when lowering the yard became a less standard means of furling the sail.

One of the problems with martnets is that they are often shown apparently lead through blocks, dumbsheaves or fairleads attached to the yard, often well out towards the end of the yard. Since the euphroe, block or whatever that joined the strands of the net together couldn't run through that fairlead, the utility of the device for bundling up the sail would be negligible.

However, this appears to be a misunderstanding. The two nets on the forward and the aft side of the sail were joined by a bridle and that bridle ran through a thimble, deadeye or dumbsheave at the end of the martnet fall—the line which went to the block below the hounds and led down to the deck. When the martnet was slacked off that thimble would tend to sit on the yard resembling a dumbsheave or block attached to the top of the yard. But it was not attached to the yard.

There seems to be general agreement in the literature that the strands of the martnet ran through deadeyes on either side of the sail joined by a bridle. There are some problems with this. If it was a standard deadeye with three holes, then there should be six strands—the artists don't always show six. Also it is sometimes shown as much smaller than the thimble or whatever where the bridle is attached to the martnet fall.



Another problem is that the artists often show two sets of martnets port and starboard on the forward or aft side of the sail when the sail is set. This should mean that four sets are shown when the sail is furled and the martnets of both sides of the sail are visible. But that is never the case. Did the artists normally see sails furled, count two sets of martnets, then draw the sails set and wrongly put both sets on whichever side of the sail was visible?

Martnets are always shown hanging in bites fairly close to the yard arm when the sail is furled. It would be necessary to take the thimble for the bridle out to near the end of the yard, either by climbing out along the yard, or while the yard was lowered, because otherwise it might foul on the robands when the sail was unfurled and the sail would not set properly. That would be less of a problem with topmast rove martnets.

Duyfken: the ship we are building

Marrianne MacKay [one of the Fremantle Duyfken Guides] kindly presented a list of questions posed by herself and Guides early in the construction:

How much timber will be used in this ship?

Approximately 300^3 m of felled oak; will convert to about 200^3 m roughly baulked timber delivered to Fremantle, of which less than 100^3 m will be incorporated in the finished ship.

Will it all be oak from Latvia?

Most of the ship's structure should be oak; which we are sourcing from Latvia. There may be times when we have to supplement with local timber. If we need specially shaped pieces such as knees and can't find them in our supply of oak. The planking **must** be oak; since the way this timber takes a bend will influence the shape of the ship. A different timber would give us a different shaped ship.

The decks and the spars will be pine. Probably Baltic pine.

Archaeological evidence suggests that blocks for the rigging could be made from any durable timber.

Is it steamed to shape?

No, steaming is a 19th century technique. In *Duyfken's* day planks were bent using direct heat from fires. This caused some burning of the timber but allowed the shipwrights to gradually shape the planks by eye for plank-first construction.

Are there hanging knees?

Yes, there are hanging knees on virtually every deck beam; also breast hooks, transom knees, a few lodging knees and many shaped timbers in the framing.

Is the wood chosen because of its shape and then shaped by the shipwright?

Yes natural curves and crooks are selected to provide the approximate shapes and they are shaped and fitted by the shipwrights.

What material was used to caulk the decks, etc?

When the original *Duyfken* was built, caulking was done using animal hair, and sphagnum moss was also used. Other vegetable fibres were not much used in Dutch shipbuilding circa 1600. We might use oakum for its durability and convenience.

Did a yacht carry a small boat, pinnace, etc for landing on expeditions — if so, where was it stowed and was it built from oak?

Yes there was a ship's boat. It was stowed below decks on a two deck ship, but took up so much space that even large ships could not operate their guns with the boat below deck. Therefore the boats had to be got out and taken in tow before engaging battle. No doubt the boats were often smashed in battle. It is most likely that the ship's boat was made of oak, although by the time *Duyfken* sailed to Australia she might have had an Indonesian teak boat.

Was there any defence against the torredo worm?

Yes, it was essential for ships operating in the tropics. The ships were covered with a layer of animal hair felt and tar, this was then covered with a thin layer of deal or pine planking (the "doubling") nailed on with

many square-headed nails. This was then covered with “white stuff”—usually some mix of resins, lime and sulphur. We will probably use a white anti-fouling paint.

What sort of metal was used and how long did it last?

Fastenings, chainplates, and metal parts of the rigging were wrought iron. Iron lasts longer than mild steel in a marine environment but ship's fastenings did not last very long. We have used bronze for some keel scarf bolts that would be very difficult to replace but we will probably have to use some galvanized fastenings for reasons of economy.

What was the rigging made from: sails, masts, spars, etc?

Sails could be made from flax or hemp cloth. The rope was hemp rope. Masts and other spars were usually Baltic pine.

Was there a crow's nest?

No, three drawings of *Duyfken* from the log of the *Gelderlandt* all show *Duyfken* with polacca rig: topsails set from the deck and sent down when not set. There are no crow's nests and the lower masts and topmasts are scarfed together as single poles rather than separate “fiddled topmasts”.

Did they carry spare wood, sails, etc. If so, where was it stored?

The first Dutch fleets to make the long voyage out to the Indies carried a large number of spare spars and it is likely that *Duyfken* carried some spare spars. All ships must have carried some spare sails which would have been stored somewhere below, and regularly checked for dampness and rodent damage.

Where were spare spars and any other long timber stored?

The deck layout of 16th-17th century ship would not provide a long weather deck where spare spars could be lashed down. Perhaps they lay along the waist between the forecastle and the quarter deck. Ships on long voyages probably looked a great deal more cluttered than they appear in artists' representations.

How much food did they carry for a long voyage. What was the diet like?

They tried to carry enough food for a long voyage. Carbohydrates were dried bread or ships biscuits, rice, flour, oats, sago from Eastern Indonesia, in fact anything that could be got. Protein was salt-preserved meat and fish, plus cheeses. Fresh vegetables and sources of vitamin C were the great problem. The role of citrus fruits, vegetables, even grass or the hearts of palm trees in preventing or curing scurvy was well known. But carrying and preserving anti-scorbutics was a seemingly insuperable problem. Perhaps the wine which they drank every day until it ran out provided some vitamin C. Palm sugar and tamarind from the Indies might do the same on the return voyage.

How much water was carried — was alcohol allowed on board?

That question could be reversed. How much alcohol was carried, was water drunk on board? Water was not drunk until the beer ran out and even then, water was almost never drunk except mixed with some form of alcohol. Voyages were of long duration, so beverages to last for several months had to be carried. Water was not a safe beverage, and after a few months stored in barrels in the hold it was putrid. Neither tea nor coffee had been introduced to Europe. Most people, including children, drank beer when they were thirsty and wine or something stronger if they felt like a drink. Scarcely anyone was completely sober except between waking and breakfast.

How was order kept? What type of punishment was carried out and by whom?

The new Dutch republic was relatively egalitarian and democratic, but order and hierarchy were enforced as far as possible. Early VOC records show that the crews on many of the ships stationed in the Indies were effectively mutinous — on occasions they did not carry out orders. Willem Jansz. was one of the few captains who did not have problems with his crews. We know that a man on *Duyfken* was keel-hauled for theft on the first voyage to the Indies. Theft on board ship was often dealt with very harshly, sometimes by the crew.

Were the sailors professional crew or were they pressed into service?

There was no legal impressment of crews. Some of the crew who signed on the 1603 VOC fleet under Stephen van der Hagen were horrified when they learnt of the war-like intentions of van der Hagen and the VOC. They had signed on as merchant mariners.

What was the pay structure like?

We don't have much information on this but many foreigners served on VOC ships because the pay and provisions were adequate.

Did the captain get a percentage of the cargo and was a bonus passed to the crew on the completion of a good voyage?

We don't know, but if the VOC did ban officers from doing any trading on their own account, the ban was probably not enforceable. There is no evidence of the crew enjoying a bonus from successful voyages — they were lucky to get home alive!

Was there a Dutch Navy and was there an interaction with the VOC fleet?

In *Duyfken*'s time there were vessels built for naval purposes but they were mainly fairly small, and intended for actions against the Spanish forces or pirates on the Zuider Zee and on the coast. There were many more armed merchant vessels built for privateering and trading to the Mediterranean and the Spanish controlled West Indies. The Dutch were

developing a navy and in 1608 they defeated the Spanish naval forces at the battle of Gibraltar. VOC ships were used as naval vessels in times of war. At the beginning of the 17th century there was little difference between armed merchantmen and purely naval vessels. East of the Cape of Good Hope the VOC had the power to make treaties and war and operated as a sovereign power with a navy.

What were the living arrangements? Was there a galley? How did they cook?

We will probably learn from the *Duyfken* replica more about what was possible and convenient in terms of messing. Presumably there were tables and benches. The galley was a simple wooden or metal box, in which a fire could be built and there were arrangements for hanging pots over the fire. Cooking was no doubt very simple. The 17th century Dutch enjoyed a similar reputation to the 20th century English in terms of culinary expertise.

With a crew of twenty, what was the watch system — how many would be required to run the ship?

Both double and triple watches are thought to have been used. Perhaps with only twenty crew, including officers, a two watch system was preferable. We will find out from sailing the replica how many crew are required for normal operations. *Duyfken* had a rig which was relatively light and simple to handle and required a relatively small crew, nevertheless her crew of twenty is a large crew for a vessel of only 25 last cargo capacity. This suggests that she was relatively large vessel for her capacity.

When sailing in convoy, would there be a supply ship carrying spare wood, sails etc. as space on small ships like *Duyfken* would be very limited?

If necessary, one ship might take spare spars or provisions from another, but all ships needed to be self-sufficient because convoys so often became dispersed in storms and, without radio or radar, regrouping was very uncertain. *Duyfken* sailed back from the Indies on her own after losing contact with the larger ships off South Africa.

As the VOC was a purely private company what was the official line from the Dutch Government regarding relationships with indigenous peoples?

Although set up with private finance, the VOC was set up by dictate of the Government and acted as the Government's proxy in all dealings east of the Cape of Good Hope. They had the authority to make war and make treaties like a sovereign nation. When the VOC was liquidated, at the end of the 18th century, the Government took the line that the VOC had acted for the Government, so its assets were the Government's. At the time of *Duyfken*'s voyage, VOC relations with South-east Asian peoples and Kingdoms were largely amicable, but little more than a

decade later the VOC was responsible for appalling genocide in the Banda islands in their efforts to gain a monopoly.

Were replacement crews available from larger ships in the convoy if required?

When Wolphert Harmensz's fleet stopped at Mauritius, an extra man was assigned to *Duyfken* because they had not been able to stand full watches.

What was the living area like — as a commercial ship was there a hierarchy — officer's quarters, etc?

Dutch ships had a reputation for being stuffy and damp. No ships of the time had many windows for ventilation and all wooden ships became very damp in bad weather. Hammocks had not yet been introduced from the Americas and the crew did not have cots. We believe they curled up to sleep anywhere that was reasonably dry and, in the tropics, they probably slept on deck. Aft, where the officers would have their accommodation, there was headroom, but it would still have been cramped.

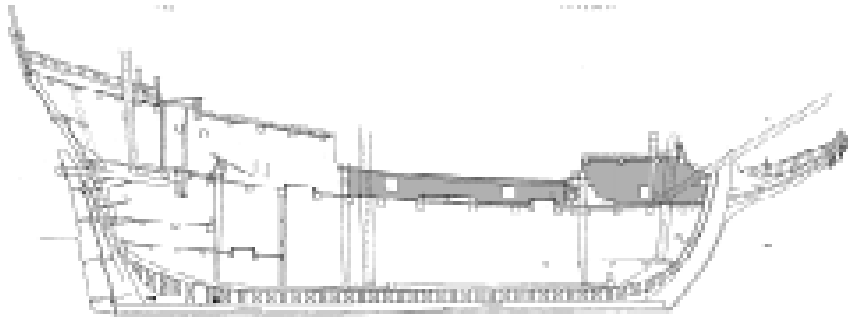
Were women ever carried on board?

We have no specific evidence that women sailed on board *Duyfken*. However, VOC ships did carry passengers of both sexes. In later centuries quite a number of women married to sailors went to sea—though their presence in ships is not much reflected in official histories—and perhaps the same was true in the early 17th century.

Did the crew have designated roles: ie bosun, sailmaker, cook, etc.

Yes, but not to the extent that was the case later. There was the skipper and his mates or officers, called "up-per" and "under steersmen". Sailmaking and rigging were less specialized than in later centuries and on a small ship such as *Duyfken* sailmaking and rigging duties were probably shared by the mates and other experienced sea-men.

Main Deck



Duyfken's main deck is the gun deck. There are four gunports on each side, but the forward gunports are only used by moving a gun forward. This would happen if *Duyfken* was chasing another ship. The gun firing for-ward would be called a bow chaser. (The guns in the stern can be called stern chasers.)

Duyfken's guns are three-pounders — they fire iron shot that weigh about 3 pounds (1.3kg), about 70mm diameter. (On board ships, the term “cannon” is reserved for a particular calibre and shape of gun — cannon were the very large guns, carried by big ships, firing 42 pound shot. Demi-cannon were smaller at 32 pounds and were the main armament of most large “line of battle ships”. There were several other sizes of gun-sakers, falcons, falconets, minions, robinets, culverines and demi-culverines, perriers, port pieces and bases. *Duyfken's* guns could be called falcons.

The gun carriages are modelled on those found on the 1628 wreck of the Swedish ship *Vasa*. Note that the trucks (wheels) fit very loosely onto the tapered axletrees. The taper of the hole in the truck causes the trucks to always track inwards and the loose fit ensures that they never jam.

In the middle of the main deck is the main hatch which has a grating to allow light and air into the hold in good weather. Cargo is loaded and unloaded through this hatch.

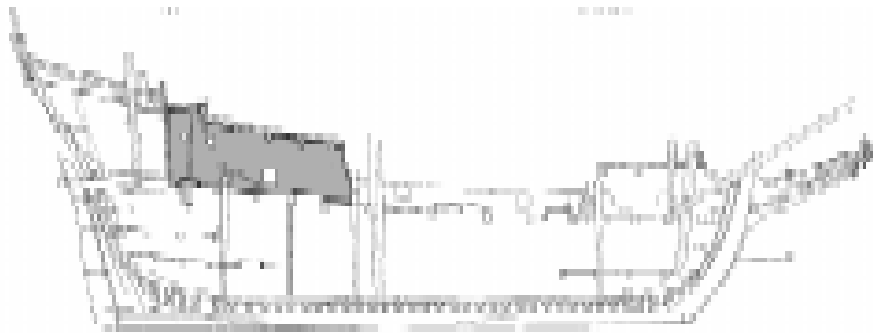
Lying across the deck, just aft of the forecastle, is the hand-spike windlass which is used for hauling up the anchors and sometimes used to lift heavy items in and out of the hold. It is operated by pulling on hand-spikes or windlass bars which are fitted into holes in the barrel of the windlass.

Also on the main deck is the galley. The galley is a simple iron box with sand in the bottom. The cook makes the cooking fire in this box and hangs the pots over the fire, very much like a camp fire. Some ships of

Duyfken's time had their galley down in the hold, but there was no chimney to take the smoke away and spilled foodstuff fell into the bilge causing a terrible stench and rotting the ship's timbers.

Around the deck should be items such as wooden staved buckets, a canvas bucket for getting water from over the side, a wooden skeet for throwing water around the decks (to prevent them from drying and cracking), and brooms for sweeping the deck.

Main Cabin



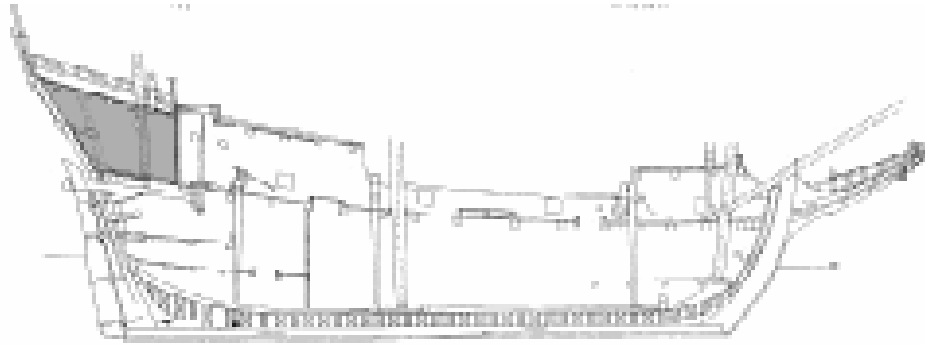
On the main deck, under the quarter deck, is the area which can be called the main cabin. There are two guns here. In the middle is the mess or table where the crew take their meals in messes of six men. The simple table is easily dismantled and moved out of the way. On the table are the wooden bowls and wooden spoons, larger serving bowls, serving spoons, and staved wooden tankards from which the seaman drank the beer or wine with each meal. Each seaman had his own bowl, spoon and tankard, he also had a knife and, if necessary he could make a new bowl, spoon, or repair a tankard using wood from the cook's supply of fire wood.

At the aft end of the main cabin is the whipstaff for steering and the steersman's position. Forward of the whipstaff is the binnacle containing the compass, the half-hour glass, and a lantern so the compass can be seen at night.

And forward of the binnacle are the two bilge-pumps. These pumps are called elm-tree pumps because the pump barrels are made from elm-tree trunks. The barrels have been bored out by hand. The pump mechanism uses two valves. The valve in the plunger sucks water into the barrel when the plunger is pulled upwards, the other valve, at the bottom of the chamber where the plunger operates, prevents the water falling back into the bilge when the plunger is depressed. It takes four men to operate both pumps effectively.

Some sea chests are stowed around the sides of the main cabin.

Captain's Cabin



Duyfken's captain, Willem Janszoon, would have shared his cabin with the merchant Jan Roosengeyn.

Dutch east India Company ships sailed for commercial gain. The trade was the responsibility of the "upper merchants", "under merchants" and "assistant merchants" who could also be termed "supercargoes" in English. These men were skilled commercial officers, possessing diplomatic finesse, cunning and determination.

But on a ship as small as Duyfken, neither the captain nor the merchant could expect very much finery or comfort.

In the cabin are two narrow bunks which are also storage lockers. The mattresses were stuffed with animal hair, not straw because straw was unhealthy in a damp environment and did not last.

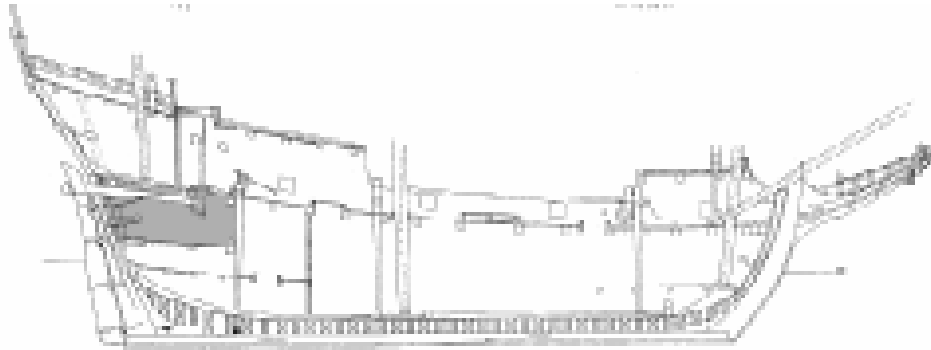
There is a bench across the stern of the ship — a standard feature. Perhaps the two most important items are the iron bound chest containing valuable trade goods and currency, and the desk where Roosengeyn could do his accounts and keep his journal, and where Janszoon, along with his upper steersman, could consult charts or draw new ones, make navigational calculations, and keep his journal.

On the desk we can see Janszoon's logbook or journal open to pages showing the first European charting of an Australian coastline. Captain Janszoon's astrolabe is also on display. This is the instrument that he uses to measure the altitude angle of the sun at midday so that he can calculate the ship's latitude.

The ink well on the desk includes a receptical for white sand which Janszoon or Roosengeyn would sprinkle over their journal pages to dry any excess ink.

Both men have pistols which can be shown to visitors.

The Constables Cabin



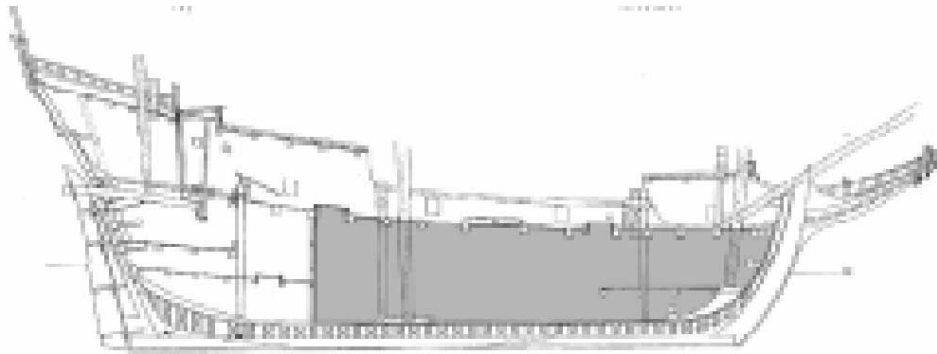
The Constable was in charge of the powder and shot in the magazine, the issue of arms on board, the operation of the guns, and also access to the ship's supplies including the beer, wine, brandy and other beverages.

Seamen were not allowed access to the supplies or the hold without supervision. This was to prevent pilferage and also because it is dark down below and anyone going there needs a candle or lantern — fire was always the great risk for wooden ships.

In the Constable's Cabin there are two cannons which take up much of the space. The tiller sweeps across just below the deckhead (the equivalent of the ceiling in a house). And there are several chests containing personal possessions, firearms, and another contains medicines and bandages, because the Constable is also surgeon on board a small ship. The Constable has labeled some medicine jars in Latin, but we are not sure that he is really an expert apothecary.

Forward of the Constable's cabin are the bread room on the port side and the magazine to starboard. Both of these sealed rooms are lined with copper so that rats can-not chew their way in and get at the ship's biscuits or the cloth for making cartridges. The metal lining also helps to keep the powder dry on a sometimes-leaky ship.

The Hold



Visitors to the ship see the hold during the process of “rummaging”. This is done before loading a new cargo. Any barrels and cargo in the hold are moved to expose the brick ballast and the ship’s timbers so that they can be cleaned. There is already some spice cargo on board packed in woven baskets. These have been piled towards the aft end of the hold, but this is not a proper stowage for going to sea.

There are barrels down in the hold, though most of the barrels have been taken ashore to collect water. Some barrels are used for dry storage of foodstuff and some contain the salted meat and fish that make shipboard food so delicious.

Duyfken Guides' Guide

The Original Ship & Historical Background

Detail from a chart in the log book of the ship Gelderlandt, showing Duyfken at anchor on the south west coast of Mauritius, October, 1601

Duyfken: A Brave Ship

Built about **1595** in the Netherlands. A fast, lightly-armed ship probably intended for small valuable cargoes or privateering.

- 1601** Selected as the jacht, or scout, for the "Moluccan Fleet" sailing to the Spice Islands.
Duyfken's captain for this voyage, Willem Cornelisz Schouten, with Le Maire, would later discover and name Cape Horn after the city of Hoorn.
On Christmas day the five ships of the Moluccan Fleet reach Bantam (Banten), Java and encounter a blockading fleet of Portuguese ships totalling eight galleons and twenty-two galleys. They engage this fleet in intermittent battle until driving them away on New Years day. This is a turning point in history: the undisputed dominance of the Iberians in the Spice Trade to Europe is over.
- 1602** Warm welcome in Bantam, repair to battle damage. Survey of Jakarta Bay, where the Dutch would later build Batavia their capital in the Indies, then sailing by way of Tuban, East Java to the Spice Island of Ternate.

Loaded cloves at Ternate then to Banda for a cargo of nutmeg. Sent on a voyage of exploration to the east.

The newly-formed United Dutch East India Company (VOC) was granted a monopoly on trade to the Spice Islands by the government of the united Netherlands.
On the voyage home from the Indies *Duyfken* was separated from the larger ships in a storm off Cape Angulhas. *Duyfken* reached the Netherlands two months ahead of the larger ships early in 1603.
- 1603** In December *Duyfken* set out on a second voyage to the Indies in the VOC fleet of Steven van der Haghen and with Willem Janszoon as skipper.
- 1604** The VOC fleet captures two Portuguese ships in Mozambique Channel and sails to the Spice Islands via India finally reaching Bantam, Java on New Years Eve.
- 1605** *Duyfken* is in the fleet that recaptures the fort of Van Verre at Ambon from the Portuguese. Later in the year she is selected for another voyage of discovery to the east, but first she is sent to Bantam Java for urgently needed provisions.

- 1606** Early in 1606 Willem Janszoon and Jan Roosengijn take *Duyfken* southeast from Banda to the Kei Islands, then along the south coast of New Guinea, skirting south of the shallow waters around False Cape and then continuing east-southeast until they reach and chart the shores of Australia's Cape York Peninsular.
- 1607** *Duyfken* may have made a second voyage east to Australia. Later in the year she is sent to Java to get supplies for the beleaguered Dutch fortress on Ternate.
- 1608** Engaged in a five hour battle with three Spanish galleys. In June *Duyfken* was sent with larger ships to capture the fortress of Taffaso on Makian Island. A month later she was brought in through the stockade at Ternate for repairs. It seems that she was hauled on her side to repair the bottom but this caused further damage and she was condemned as unrepairable, and was left to the probable ignominy of being used as construction timber and even firewood.

The Whole Story

The whole story should obviously start somewhere near the beginning.

Australia separated from the super-continent of Gondwanaland a very long time ago. Separate evolution produced a unique mammal fauna. At sometime, probably, between about 100 000 and 40 000 years ago humankind first reached Australia.

At times during the last 100 000 years Australia has been joined to New Guinea by a land bridge, but, contrary to a widely held belief, Australia has never been joined to Asia. In fact the Australasian continent is closer to Asia now than at any time in the past and we are moving northwest at about a metre a century.

The ancestors of the Aboriginal peoples were seafarers. We do not know what kind of watercraft they used but bamboo rafts are widely fancied as the most likely type.

During recent pre-history there were probably other visitors to Australia. The dingo was introduced about 4-5000 years ago from Southeast Asia. The Malayo-Polynesian peoples are the first to have crossed wide seas and oceans so there is no difficulty with accepting that they made the short trip to Australia on many occasions.

Who else might have visited? The Chinese had a brief period of maritime exploration and expansion in the southern Ming (early 15th century) and might have visited Australia. Arab seafarers might also, but any apparent knowledge of a southern continent on the part of Arab or Chinese navigators was most likely gained from Southeast Asian pilots who they employed.

There are claims for Portuguese, Spanish, and French discovery of Australia during the 16th century. The evidence for these discoveries does not stand up to scrutiny and some of it is clearly the work of cranks.

Duyfken's voyage to Australia in 1606 is the first voy-age of which there is historical evidence.

Duyfken's captain Willem Janszoon surveyed and charted a portion of the Australian coast.

For the first time ever it was possible for all the World's inhabited continents to be drawn on a map.

Two-hundred years earlier map-making (cartography) was a very imprecise science and theoretical geography was based largely on Greco-Roman authorities and biblical Scripture. There was very little expectation that going out and exploring the World would add much to the science of geography. This was typical of the Medieval World View. The Age of Discovery, the 15th and 16th centuries is also the renaissance, the time when Europeans took a huge leap forward and began modern ideas of scientific inquiry and reason. The discoveries of the great navigators were very important in validating the idea of scientific inquiry.

Cartography and navigation, practical sciences of the mariner, had improved to the point where exploring navigators could accurately record what they found: this was hugely destructive of theoretical geography.

A World map of 1400 typically shows Jerusalem at its centre and four symmetrically arranged continents around it (Europe, Asia, India and Africa.)

By the end of the 16th century, the World map included the outlines of the Americas, Europe, Asia and Africa in recognizable form. Theoretical geography still imposed a huge Southern continent to counter-balance the northern continents. *Duyfken's* voyage did not immediately dispel that theory, but it was the beginning of the transformation of Terra Australis Incognita to the Nation of Australia.

The Historic Background to Duyfken's Voyage

When *Duyfken* sailed east from the Spice Islands in 1606 she was owned and operated by the Dutch United East India Company (VOC Verenigde Oost-Indische Compagnie). A State set-up monopoly to control trade to and from Asia.

The VOC had remarkable powers: it could raise armies, wage war and make treaties on behalf of the Netherlands. No other European Government would have given such powers to a commercial organization; but the Netherlands were a Merchant Republic not a monarchy.

The Netherlands or Low Countries were newly independent. They had been ruled by the Spanish Crown through the Dukedoms of Burgundy and Parma.

The Protestant Reformation in northern Europe and the counter attack of the Roman Church, at its most repressive in the Spanish Inquisition, made the region less and less amenable to Spanish backed administration.

That Administration collapsed in 1565 and the “Low Nobility” of the region took over for a while requesting relaxation of edicts and taxes against Protestants. In Antwerp the cathedral was sacked and many sacred relics destroyed in 1566.

King Philip of Spain sent the Duke of Alba to rule in the Lowlands in 1567. He did not make himself very popular. The Dutch War of Independence — the “Eighty Years War” — is considered to have started in 1568.

In 1572 Prince Willem of Orange was negotiating assistance from Protestant Germany, England and Huguenot France intending to attack Spanish forces in the Netherlands in the summer.

It was a loose alliance of sea traders, fisherman and pirates or privateers (the same ships and sailors were often involved in all of those enterprises) who struck the first blow capturing the town of Brielle on 1 April 1572 and going on to take Flushing, Veere and Enkhuizen almost immediately and taking Oudewater, Gouda, Dordrecht, Leiden, Hoorn and Haarlem later in the year. Not all of these cities stayed long in rebel hands. Haarlem fell to a Spanish Siege in July 1573, but the siege of Leiden was eventually broken by sea forces.

By 1576 the Spanish recognized that they were fighting a losing battle in the Protestant northern Netherlands and a truce was negotiated. The 1579 Union of Utrecht formed the first United Provinces of Netherlands consisting, in principle, of both the northern provinces and the Catholic southern provinces. It was a republic, but the Royal House of Orange was the most powerful force in the Republic, and was regarded as leading the country by European monarchies. Philip of Spain offered a reward for Prince Willem’s assassination.

A formal declaration of Independence (Abjuration) from Spain was declared in 1581.

Prince Willem was assassinated in 1584 and the seat of Government was moved from Brussels to the Hague: this had the effect splitting the Catholic south from the Protestant (Calvinist) north. The seven northern provinces Holland, Zeeland, Utrecht, Gelderland, Overijssel, Friesland and Groningen formed the nucleus of the present-day Netherlands. Rich and educated Catholics moved from the Spanish dominated lowlands (now Belgium) to the north giving much of the impetus to the Dutch Golden Age of Commerce, Science and Art in the following century.

Dutch Shipping Circa 1600

The Netherlands was a small area of land but virtually controlled the shipping of northern Europe and was increasingly dominant in shipping to Spain and the Mediterranean.

The herring fishery was the foundation of the Dutch maritime wealth. The fishing vessels fished during the winter months and during the other six months of the year carried cargoes. They were as large as the average unarmed cargo ships of the time and were of an efficient design.

During the course of the 16th and 17th centuries England lost more and more cargoes to Dutch shipping. Various reports to parliament at the time discuss the reason for this and it was universally agreed that Dutch ships carried far more cargo for their size and sailed well.

English historians have attempted to explain this by saying that the English ships were more heavily built and carried heavy cannons. But English shipping to the Baltic, like Dutch shipping, was generally unarmed or lightly armed.

On both sides of the North Sea there had been a transition from building clinker (over-lapping planks) Medieval ships to the three-masted galleon type built with flush (carvel planking). Both the English and the Dutch were, to some extent, copying Mediterranean and Iberian design but they did it in different ways.

The English (in the Naval dockyards at least) adopted the Iberian system of frame first ship construction. The mathematics and geometry used to determine the shape of the frames was not very sophisticated and so the shape of ships was constrained to a range of less-than-optimal designs.

The Dutch continued to build plank first, shaping the ship by eye, and giving it the shape they felt would serve best. Instead of using the

overlaps of the planks to join them together temporary cleats were used until the frames were inserted.

Dutch ships could carry a lot more for their size, sailed efficiently and required a relatively small crew.

Their enemies the Spanish tried to ban Dutch ship-ping from their ports. And the United Provinces tried to ban their merchants from trading with the enemy. But the Dutch carried so much of the trade in bulk commodities that such a ban could not be imposed. The Spanish would have starved without the Baltic grain imported in Dutch ships. The Dutch merchants would have had a disastrous number of unemployed ships without the trade to rich Spain and Portugal.

Spain which controlled Portugal in the late 16th century increasingly excluded Dutch traders from the lucrative Spice Trade. Inevitably traders of the confident and ambitious new nation decided to go and get the spices themselves.

Geographical theory suggested that the Orient could be reached by sailing north around Asia. Willem Barentszoon made three attempts to sail north around Russia. On the third voyage his ship became trapped in the ice and he and the crew had to spend a terrible autumn, winter and spring on the island of Nova Zembla. Bartentszoon did not survive the experience though most of the crew did.

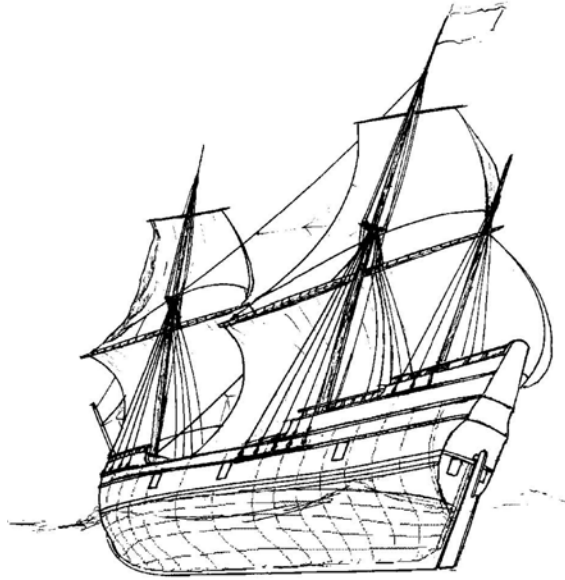
The first Dutch voyage following the Portuguese route around the Cape of Good Hope was made in 1595–7 lead by Cornelius de Houtman who did not do well in negotiating with the Raja and traders at Bantam Java. His fleet included a ship called *Duyfken*.

A second voyage followed in 1598 led by van Neck. Four of his eight ships loaded at Bantam Java and re-turned the following year. A round voyage of fifteen months which made a wonderful profit. The *Duyfken* of the first fleet, renamed *Overijssel*, sailed with van Neck's fleet.

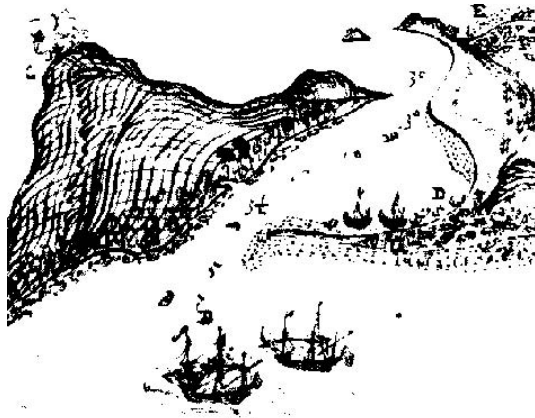
The rest of the fleet split in two and went to Banda and Ternate in the Spice islands. The first Dutch ships to load at the source of the Spices rather than the entrepot of Bantam.

Duyfken: a short history

Harmenszoon's fleet went to the Spice islands rather than loading at Bantam. Three ships went to Ternate, the source of cloves, while the other two went to Banda where the Phantom lives and where all nutmeg and mace came from. There was not much cloves to be had, so the three ships *Gelderlandt*, *Wachter* and *Duyfken* sailed down to join the other two at Banda and load nutmeg and mace.



While they were waiting for cargo, *Duyfken* was sent off exploring to the east for the first time, asking about the island of Nova Guinea where gold was supposed to be found in great store and buying sago for provisions on the return voyage.



Duyfken at Banda

Back in Banda the flagship *Gelderland*, and *Utrecht* were laden. *Duyfken*'s small hold was soon filled and the three ships set off for Java and the Netherlands.

Zeelandia and *Wachter* remained in Banda waiting for cargoes. *Wachter* went missing on the return voyage.

Meanwhile, back in the Netherlands, the Government reacted to a problem in the spice trade: the glut of spices reducing prices in Europe and greatly increased demand forcing prices up in the Indies. The

United Dutch East India Company (VOC) was granted a monopoly on trade to the Indies.

The new company bought *Duyfken* from the Oude Compagnie before she returned to the Netherlands.

Duyfken and *Utrecht* were separated from the *Gelderland* when her sails were blown out at night in a storm. *Duyfken* was sent ahead to search for *Gelderland* (which was actually lagging behind) and arrived home in Feb 1602, two months ahead of the larger ship.

Ten months later she sailed for the Indies again in a large VOC fleet commanded by Stephen van de Hagen and with Willem Janszoon as her captain. She was intended to remain in the Indies.

In 1605 she was in the fleet which recaptured the Dutch Fort van de Verre at Ambon in the Spice Islands and was then selected for another voyage of exploration to the east.

In June of 1606 her return was reported to an English merchant in Bantam, John Saris, by the captain of a Javanese trading ship that had been in Banda.

The following year the VOC merchant Paulus van Solt arrived in Ambon and recorded in his journal that he found at Ambon “the jacht Enckhuysen full with cloves, and the jacht *Duyfken*, that had come from Nova Guinea”. This seems to suggest that *Duyfken* had made a second voyage eastwards in early 1607. Nova Guinea in this context is not our New Guinea but a semi-mythical source of gold that was either part of Terra Australia Incognita — the great southern land — or a large island lying just off its coast.

At this time Portuguese resistance to the Dutch incursion into the Spice trade was belatedly hotting up. *Duyfken* was amongst the ships sent on an aborted attempt to capture the Portuguese fort on Tidore. Instead they set-up a new fort at Malayo on neighbouring Ternate and *Duyfken* was stationed there. She was sent to Bantam to get supplies for the garrison.

While at Bantam she underwent some “very necessary repairs”. A full cargo of supplies could not be got at Bantam so she completed her load at Gresik (near Surabaya, East Java) and finally returned to Ternate on 20 January 1608. The commander of naval forces there, Jan Roossengin (formerly *Duyfken*’s supercargo who had sailed to Australia) wrote that the “rice, beans and a bit of arak ... brought much joy amongst our people”. The bit of arak was actually seven very large barrels (*leggers*).

The Portuguese fort on the other side of the island was being strengthened by the arrival of more ships from Manilla.

Duyfken and another jacht were employed harassing small ships, pirating Asian shipping that traded with the Portuguese and trying to lure the larger Portuguese and Spanish ships out to sea to do battle away from the protection of their shore battery.

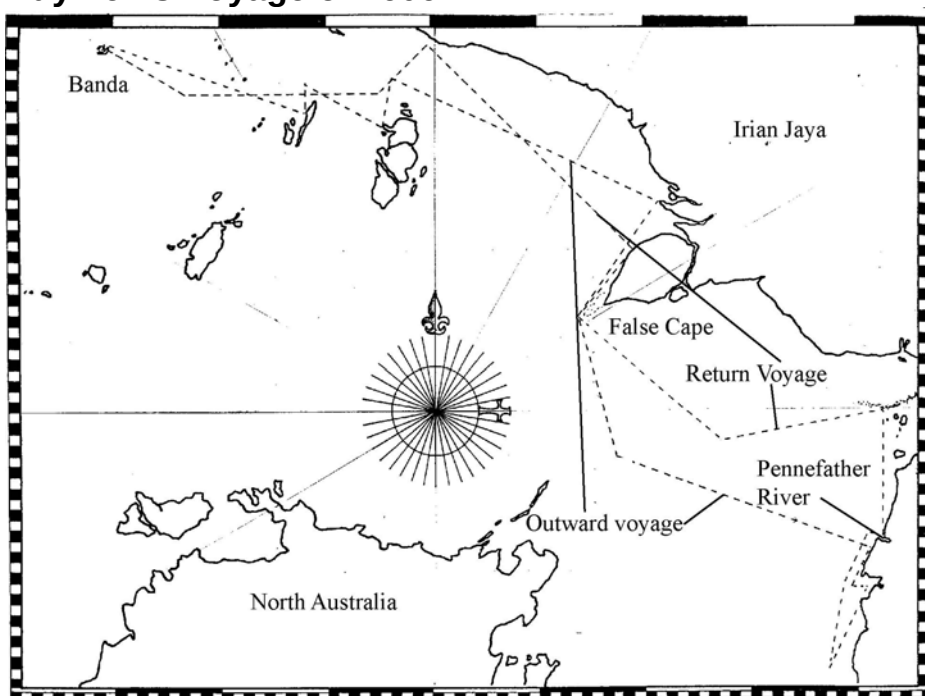
In June, an unsuccessful attack was made on the Portuguese at Tidore, then the fort on the island of Makian was successfully taken, but two of the Dutch ships were wrecked whilst lying at anchor at the roadstead in front of the fort. The Dutch decided to move their fort to the other side of the island.

Duyfken was sent back to Ternate and was hauled in through the reef for some more repairs. She was then hauled over on one side (careened) so that her bottom could be worked on, but this caused further damage and she was condemned as un-repairable on the beach at Ternate in July 1608.

Probably she was fairly lightly built, she had been hard worked and years in the humid tropical conditions had doubtless rotted many of her upper timbers.

Most likely she was broken up and her sound timbers used in constructing the new fortress while small and rotted pieces found their way to the kitchen to fuel the stoves.

Duyfken's Voyage of 1606



In search of the real Willem Janszoon

Pieter Both, a VOC official wrote about Willem Janszoon:

“by nature a moderate, simple and modest man, who had acquitted himself excellently in his task ... and had never had any trouble with his crew and had always managed to maintain order on board.” The man who captained *Duyfken* on her historic voyage to Australia was a loyal and respected servant of the Dutch United East India Company.

Like James Cook he seems to have been a stolid and dependable man. He was surrounded by more colourful, though less reliable, characters such as the demoted steersman Jooris Joustenszoon who drew the sketches of *Duyfken* in the *Gelderlandt* log book and waggishly sketched the Admiral at his toilet on the stern gallery lowering a bucket to get water. Little of Janszoon's personality reaches us through the archived documents.

Early career

There is some question about Janszoon's early career. Willem Jans, Jansz or Janszoon is about the most common name in the Netherlands circa 1600, so it is difficult to tell whether archival mention of Willem Jansz refers to the man who skippered *Duyfken* in 1606. A Willem Jansz was first mate on the ship *Hollandia* sailing to the Indies in 1598. A Willem Jansz of Amsterdam was mate of the *Gouda* sailing to the Indies in 1600 and another Willem Jansz was skipper of the *jacht Lam*, one of three vessels, *Ram*, *Lam* and *Schaap*, of the *Compagnie de Moucheron*, commanded by Spilbergen sailing to the Indies in 1601. Second in command Guyon le Fort, who had been to the Indies before, was accused of conspiring to mutiny and removed from his command. The ship *Ram* was decrepit and was sold never to return to the Netherlands.

Lam and *Schaap* slowly loaded pepper in Aceh and Bantam until September 1603 and reached Vlissingen 24 March 1604. Mathias Smit was the skipper of *Lam* on the return voyage. Janszoon could not have been on the *Lam* if he was the Willem Janszoon who skippered *Duyfken* since *Duyfken* had departed for the Indies three months previous. Either these are two different Willem Janszoons or Janszoon had returned earlier. Was he implicated in le Fort's mutinous conspiracy? Our Willem Janszoon was *Duyfken*'s skipper when she sailed to the Indies in 1603 with Steven van de Haghen's fleet, and he remained with *Duyfken* until sometime after the 1606 voyage to Australia.

After *Duyfken*

In October 1606, Saris at Bantam noted the arrival of the ship *West Friesland* from Ternate. It is not clear whether Willem Janszoon was then the skipper, but he certainly was in December when she sailed for the Netherlands with two other ships. *West Friesland* was wrecked on a sandbank somewhere off Mauritius. The crew were saved but little of

their possessions or the cargo were salvaged. Janszoon made his way back to Java in a ship's boat. He was exonerated of blame for the loss of *West Friesland*. Wages that had been withheld were eventually paid about ten years later.

In 1608, Willem Jansz was asked to make a voyage to China where, it is noted, he had been before. Therefore, it seems he is the Jansz who sailed to China on the *Gouda* in 1601 and made a chart and set of sailing directions for the east coast of Malaysia and Thailand.

In following years, our Willem Jansz skips a number of vessels and voyages to Aceh and Ceylon on diplomatic missions. In 1611 he returns to the Netherlands, but within a year or two he sails again to the Indies as skipper and merchant on the ship *Oranje*.

In 1615 he was appointed Governor of the Island of Solor. Although never of great historical significance, Solor was a fortress with a colourful history. Janszoon was certainly not the most colourful of its Governors, indeed he was one of the few officials who did not convert to Catholicism, join the Portuguese, and make difficulties for the VOC.

The Portuguese built the fort on Solor during the 16th century, and it was often threatened by local rebellion or Javanese and Ternatean opposition. Sometimes it was damaged by firing of the thatched roofs and sometimes by earthquakes. The fort of Solor was positioned to command the Solor and Flores straits, and therefore to control the sailing route to Timor and the sandalwood trade.

The Raja of Buton is believed to have sent an Englishman in his employ to Ambon to persuade the Dutch to attack Solor because this would alter the power balance to prevent Macassar/Gowa from attacking him. A siege and attack in early 1613 were eventually successful in taking the fort.

Dutch role formulated

In January 1614 Governor General Pieter Both formulated the Dutch role for Solor. Frigates based there would intercept all Timor (sandalwood) trade by Indonesian and Chinese traders. With a monopoly of sandalwood the Dutch could break into the lucrative Chinese silk trade with a commodity that the Chinese valued. European traders always faced the problem that they had very little to offer that was valued in Asia. Woolen cloth was not attractive to people who had silk and cotton. Most things of value — spices, porcelain, gems — came from Asia.

Adriaen van der Velde, the first Dutch commander of Solor found it difficult to prevent the Chinese trading since "They can give more than we, for in China everything is abundant and cheap" and he complained that the VOC monopoly was causing ruin, "whatever place the VOC

took over in the Indies necessarily decayed and declined” he observed. Van der Velde requested missionaries. He got the cantankerous Mathys van den Broeke who had already failed through arrogance in Ambon. Two years later van der Velde was killed when trying to suppress a Christian revolt on the neighbouring island of Adonara. The pro-vost, Joost van Malen, then deserted to the Portuguese at Larentuka and encouraged them to attack the fort.

Unpopular monopoly

Willem Janszoon replaced van der Velde. He seems to have sorted things out, but perhaps he did not rigorously enforce the Company’s unpopular monopoly.

The VOC sent a delegation to see whether the Solor fort could be made profitable and they concluded that the monopoly could be enforced using ships — there was no need for the fort. But Governor General Reynst decided to keep the fort operating to see whether the quantity of sandalwood levied from Solor could be in-creased. In reality, the straits commanded by Solor were only one of the possible sailing routes to Timor and the Asian shipping simply went by another route. At the end of 1615 Reynst ordered Janszoon to raze the fort and abandon it.

A few years later the Dutch returned and repaired the fort. Jan Thomaszoon Dayman was appointed commander in 1622. Like the provost before him, he converted to Catholicism and deserted to Larentuka with a Solorese wife in 1625. His successor, Jan de Hornay, said to be debauched and drunk, gathered no sandalwood at all for the company. At the end of 1628 Gregorius Corneliszoon was sent to again raze the fort. But before that happened de Hornay converted and went to Larentuka with his Florenese wife. His son became com-mander of the Portuguese forces on Flores and Timor. He married the daughter of the Raja of Ambeno, Timor, and founded a dynasty of Topasses or “black Portuguese” who formed a third, independent, force in the region.

Meanwhile, Janszoon had gone from Solor to the new VOC headquarters in Batavia (Jakarta). He was briefed to lead a voyage to Surat, India, but the English and the Sundanese from Bantam allied to attack and besiege Batavia. Janszoon and the new Governor General, Coen, retreated to the Moluccas to regroup and in 1619 returned to break the siege. Janszoon then set out to take the battle to the English. His fleet found and captured four English ships on the coast of Sumatera.

However, the VOC directors soon put an end to the hostilities against the English, and order the Dutch and English companies to cooperate against the Spanish and Portuguese. A combined fleet was sent to Manila, commanded by an English Admiral and with Janszoon as second in command.

Cooperation between the English and Dutch elements of the fleet was not particularly good and must have called on all of Janszoon's noted diplomatic skills. At Manila, as had been agreed, Janszoon took overall command. There was no major battle, but the combined fleet succeeded in disrupting the trade to and from Manila and captured some Chinese ships. Janszoon and his fleet returned to Bantam, Java, in December 1622, sailing by way of Japan and China.

Soon after this comes the most controversial part of Janszoon's career. He was appointed Governor of the Banda Islands. These islands, which were the source of all the World's nutmeg and mace, had been entirely cleared of the indigenous population. The Bandanese had refused to cooperate with the monopoly that the Dutch wished to impose. Governor General Coen had responded by sending in the soldiers and massacring the population. Those who survived fled to other parts of the Moluccas and some of these fugitive Bandanese communities still survive in remote locations.

The Bandanese population was replaced by labourers or slaves, some of Chinese origin, brought from Java and other parts of the archipelago. Janszoon was not involved in the genocide, but the description of his work during his three years there as "agricultural reform" glosses over a shameful episode.

After setting up an efficient and profitable administration at Banda, Janszoon voyaged to Surat and Persia, and on his return to Batavia, found the settlement under siege again, this time by the forces of the Javanese Kingdom of Mataram. Janszoon commanded some of the actions which lifted the siege and with peace restored, he set sail for the Netherlands in command of a fleet of five ships. He returned to the Netherlands on July 9, 1629, presumably to enjoy a comfortable retirement.

We do not know whether he was married

Against the soap-opera history of Solor, and the infamy of Coen's rule, Janszoon appears a pale ghost. The lack of historical and personal detail about much of his career probably reflects the dictum that "no news was good news" for Janszoon's contemporaries.

Willem Jansz, wife to a Willem Jansz of Amsterdam, gave birth to a son, Seebear, van Nieuwelent, off the western Australian coast aboard the ship *Leyden* on 27 July, 1623. This is the first historically recorded birth in what is now the territorial waters of Australia. Unfortunately, the Willem Jansz is described as a naval cadet or midshipman, and cannot be the man who skippered *Duyfken* on her historic voyage.

Duyfken Myths

The history we think we know always contains a mix of factual knowledge and popular myth. For example, Captain James Cook is widely known to have discovered both Australia and the cure for scurvy, although neither is factually correct.

Most text books say that the *Duyfken* which made the voyage to Australia in 1606 was the same *Duyfken* as the one which made the first Dutch voyage to the Indies in 1595 as part of Houtman's fleet. But that *Duyfken* was renamed *Overijsell* when she joined the next Dutch fleet going to the Indies in 1598.

Overijsell is the name of one of the six provinces or states that formed the original Republic of United Provinces of the Netherlands. The other ships in the fleet were all named after the other provinces except the flagship *Mauritius* named after the head of state, Prince Maurits and the vice-Admiral's ship named *Amsterdam*. The other ships were: *Hollandia*, *Zeelandia*, *Utrecht*, *Gelderlandt*, and *Friesland*. (A seventh province, *Groningen*, joined the union in 1594.)

Overijsell continued to sail under that name and made a third voyage to Indies, arriving at Bantam on 1st September, 1600. She was still in the Indies when our *Duyfken* set out from the Netherlands on her first voyage to the Indies on 23rd April, 1601.

Our *Duyfken* cannot be the ship that sailed with Houtman in 1595. Most text books say that *Duyfken* sailed along the coast of New Guinea, across Torres Strait and down the Cape York coast to Cape Keerweer before returning to the Indies. This reverses the real direction of the 1606 voyage as shown on the *Duyfken* chart.

Duyfken sailed out into the Arafura Sea from Dolak Island, Irian Jaya, then sailed east-south-east in the open sea until making a landfall at the Pennefather River on the west coast of Cape York peninsula. From Pennefather River, *Duyfken* first sailed south surveying the coast until turning round at Cape Keerweer. She then sailed back up the coast surveying all the way to Torres Strait, then across the Strait until reaching a barrier of reef on the New Guinea side. From there *Duyfken* turned west and returned to Dolak Island and from there to Banda in the Indies.

It is widely believed that Captain Willem Janszon of the *Duyfken* believed that Cape York Peninsula was joined to New Guinea. However, the *Duyfken* chart does not show Cape York joined to New Guinea, neither do most Dutch maps produced during the 17th century. Both Carstenszon and Tasman were sent by the VOC to investigate the strait between Cape York and New Guinea, but they were dissuaded from their course by the maze of reefs and islands and treacherous currents.

Duyfken 1607 voyage to Australia?

Duyfken's historic voyage of 1606 is well documented. She was sent on a voyage of exploration to search for "Nova Guinea and other east and south lands". Nova Guinea was a theoretical land where gold was believed to be abundant, it was part of the hypothetical Great Southern continent, not the island we now call New Guinea which was already partly charted.

A chart from the so-called "Secret Atlas" of the Dutch East India Company shows the course of *Duyfken*'s voyage of exploration. The chart is a composite of three or more original charts made on voyages of exploration. The inscription on the "Duyfken chart" reads:

"This sea chart shows the way, both in sailing out-ward and back followed by the yacht Duijfien when visiting the lands east of Banda unto Nova Guinea, but guliguli on Cenam and Ceran etc. is according to the drafting of Abraham Francken A°..1602 [on the *Duyfken*], done the 20 April, and Nieuw Zelandt with the gownongapi east of there has been sailed to by Jas-per Janssen de Jonge."

It does not say in which year the lands "east of Banda and Nova Guinea" were visited. It is possible that *Duyfken* made a second voyage to Nova Guinea (Cape York Peninsula) in 1607, and perhaps that is when most of the discoveries were made.

In the Journal of Paulus van Stolt, upper merchant under Admiral Steven van der Haghen, the entry for 4th March 1607 reads: "arrived, thank God, at the castle [of Ambon] . . . found here the yacht *Enckhuysen* full with cloves, and the yacht *Duyfken*, that had come from Nova Guinea".

We know that *Duyfken* was back in Banda in May of 1606. Would van Stolt say that *Duyfken* "had come from Nova Guinea" if she had returned nearly a year earlier? It seems possible that *Duyfken* had voyaged across the Arafura Sea and Gulf of Carpentaria for a second time and recently returned when van Stolt saw her in Ambon.

What did Jansz. mean when he wrote *nova guinea* on his chart of a section of the Australian coast?

The name Nova Guinea obviously implies some comparison with Guinea in sub-Saharan West Africa. Guinea was known as a great source of gold. There must have been an idea that Nova Guinea would also be a source of much gold.

This stems from what Jack-Hinton (1969) called "The Ophirian Conjecture". The islands of Ophir and Tharshish, were the source of King Solomon's gold, according to the Old Testament (Kings I: 9–10.), and their discovery was a matter of considerable interest to the Spanish

and Portuguese for whom the Pope had divided the World into two hemispheres for their respective discoveries and exploitation. The eastern boundary of the two hemispheres, 135° E, lay close to the Spice Islands and, in the absence of any accurate means of determining longitude, there was question and debate as to which hemisphere the Spice Islands lay in. It was conjectured, and seemingly widely accepted, that Ophir and Tharshish lay to the southeast of the Spice Islands: Spain argued that the three-year voyage of Solomon's ships to Ophir and Tharshish demonstrated how very far east the Moluccas or Spice Islands were, and that this clearly put them in Spain's hemisphere (Jack-Hinton 1969:2– 3).

The location of Ophir was connected with belief in a hypothetical southern continent that had been a part of geographic theory since long before Ptolemy put it on a map, and which theoretically balanced the land mass of the known-World's northern hemisphere.

Nova Guinea was another name for this fabled source of gold lying to the southeast of the Moluccas and, it was thought, close to the "Islands of Solomon" or Solomon Islands as we know them today. Janszoon did not imply that his *Nova Guinea* was joined to modern New Guinea since the section of that island which he charted, he labeled *Os Papuas*.

Ortelius' world map of 1570 gives position and shape to *Noua Guinea*. It lies to the east, separate from the "birds-head" end of Irian Jaya (Janszoon and earlier navigators' *Os Papuas*) and stretches to about 21° S in a large promontory between 146° and 152° E, approximately North Queensland to the Solomon islands on a modern map. It was shown separated from the great southern continent (a massively expanded Antarctica) by a narrow strait but an inscription states that it is not certain whether *Nova Guinea* is part of *Terra Australis* (map and inscription reproduced in Jack-Hinton 1969:8–9). Janszoon must have been familiar with Ortelius' map, or maps with information derived from it. The coast of Cape York Peninsular that he explored from about 13° 45' S to 9° S in about 142° E (he had no accurate means of knowing the longitude) was positioned more-or-less exactly where the west coast of Ortelius' *Noua Guinea* lay. For Janszoon, that land was not part of what we call New Guinea and was probably part of a southern continent.

Janszoon was not confused about what he charted and Saris was reflecting very widely accepted knowledge, and perhaps some personal scepticism, when he wrote about *Duyfken* in 1605: *heere departed a small pinasse of the Flemmings, for the discovery of the Island called Nova ginnea, which, as it is said, affordeth great store of Gold.* Jack-Hinton, C., 1969. *The Search for the Islands of Solomon*.

The “Orange Book” Handbook for Duyfken Guides, produced during construction of the ship.

Outline:

Early in 1606, during the tropical wet-season, the Dutch jacht, *Duyfken*, under the command of Captain Willem Janszoon, reached the coast of the fabled continent *Terra Australis Incognita*. Janszoon and his men started the process of exploring, describing and charting the land which would eventually become known as Australia. *Duyfken* was the first recorded European vessel to visit here, thus becoming Australia’s “first ship”.

On January 11th 1997, HRH Willem Alexander, Prince of Orange, ceremonially laid the keel of the replica *Duyfken*. *Duyfken* was built by the *Duyfken* 1606 Replica Foundation. Supported by the Western Australian Government, the Maritime Museum of Western Australia, the Lotteries Commission, and private sponsors in the Dutch, Indonesian and Australian communities. Building took place the *Lotteries Duyfken Village Ship-yard*, within the grounds of the Maritime Museum in Fremantle. The project was intended to engender a wider understanding of the role of the Dutch navigators in Australia’s history; strengthen the historic and cultural ties between Indonesia, the Netherlands and Australia.

Replica or Reconstruction

Strictly speaking, because no original ship or plan ex-ists, *Duyfken* is more properly described as a reconstruction. The design has been reconstructed from archival and archaeological research, study of contemporary marine art, and computer modelling. Morphometric analysis of iconography, a research technique pioneered in Australia by the project’s Nautical Archaeologist, Nick Burningham, examines contemporary Dutch drawings and paintings of ships of *Duyfken*’s time using statistical techniques to avoid conclusions based on the researcher’s’ preconceptions. The *Maxsurf* computer application was used to test the performance of proposed designs. It was intended that the reconstruction would perform to the standards recorded for the original, with a maximum speed of at least 8 knots. *Duyfken*’s design is that of a small, fast, armed merchantman or privateer.

She was reconstructed using the same materials, methods, and in many cases, the same tools as were used on the original ship.

The Ship

As far as possible the replica was built and is sailed the same way as the original. The following descriptions of building, sailing and shipboard conditions include data for the original *Duyfken*, and with the exception of such things as navigation techniques, discipline, health etc., may be taken as broad guidelines for the replica.

Masts, yards and bowsprit are Maritime pine (*Pinus pinaster*). Obtained from local plantations and grown from seeds brought from Portugal. There are three masts:- fore, main and mizzen, plus the bowsprit. Fore and mainmast carry two square sails each, the mizzen carries a triangular lateen sail. It is possible to rig a small square spritsail on the bowsprit.

The sails and ropes are of hemp, as were the originals. Hemp and flax were the fibres in everyday use in those days. The seams are “paid” with a modern com-pound rather than pitch. Sail area is about 290 m, (or 3100 ft). There are about 155 blocks, deadeyes and euphroes and about 3.5 kilometres of rope used on *Duyfken*. A “polacca” rig allows the top and lower yards to be lowered to the deck for the furling of sails when required.

The hull is of Latvian oak, (*Quercus robur*). Oak from the Baltic was probably used in the original *Duyfken*. Whilst technically she is single deck vessel, she actually has decks on several levels:- main or gundeck, aft-peak, quarterdeck, forecastle deck, and poop deck. In the middle of the ship, depth in the hold is 2.4m.

Duyfken is of plank first construction. In this method, the bending of the planks gives the ship its shape, hence the importance of using the same wood as the original. Planks are fire bent by eye in the shipyard. Fastenings include bronze and steel bolts, but mainly 25mm (1 inch) oak trunnels are used. Some of the trunnels were turned by hand on a “trunnel moot”, by visiting school children, who then sign them. These trunnels are driven into the ship as a permanent link between the children and the *Duyfken*.

A ship of *Duyfken*’s size would have had at least three or four anchors and possibly a number of spares. Our ship will have two replica anchors, but will have a number of modern, lighter types for actual use. Anchors were slung on the either side of the bow when coasting, and stowed below decks when at sea. Anchor cables were hauled up using the horizontal windlass situated in the forecastle, the cable or hawse entering the forecastle through the hawse-pipes. When at anchor the anchor cable was made fast to the bitts beside the windlass, and when at sea the hawse-pipes could be stopped to pre-vent water entering. Anchors could be carried out and retrieved by the ship’s boat, which had a small windlass on board. The ship’s windlass can also be used to assist in raising and lowering yards, guns, cargo, ship’s boat etc.

Carving of the ship’s ornamentation was done by Jenny Scrayen, who was also the woodcarver of the *Endeavour* Replica. Merchant ships in paintings and drawings of *Duyfken*’s time were sometimes decorated with carving and painted patterns. The Dutch, however, used less carving on their ships than the Spanish and English. It is believed that

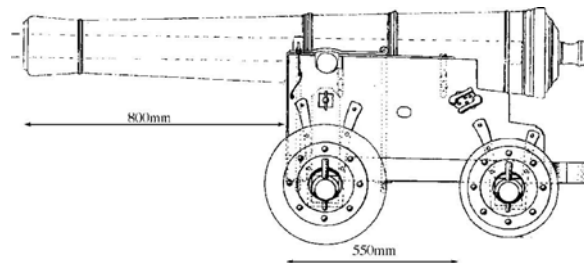
Duyfken had a stern carving depicting a dove, (her name), and the coat of arms of the city she came from. Our *Duyfken* is to have these decorations including the arms of the City of Fremantle.

Steering is by whipstaff connected to the tiller, operated by the steersman or helmsman in the hutch below the break of the poop deck. The original steersman had a binnacle which contained a compass, sandglass and lamp, our binnacle has modern versions of these devices. The steersman could look out from his hutch opening between the decks and steer the ship by compass and by watching the sails. When maneuvering, steering instructions could be relayed to the steersman by a person on the quarterdeck who conned the ship.

Forward of the steersman are the bilge pumps, using a lever operated plunger within a hollow wooden tube. The pump discharge runs down a pump dale to scup-pers. The replica has modern backup pumps.

Ordnance

Based on research, the ship's guns are approximately 3 pounders; 2 of bronze, 6 of iron and 4 small iron anti-personnel swivel guns. On long voyages the guns are carried in the hold as was normal in *Duyfken's* time.



The powder and shot magazine was in the stern below the waterline. This area was traditionally overseen by a constable who made up and issued the powder charges from gunpowder barrels. Because of the close confines below deck, the guns were probably not allowed to recoil, and had to be hauled inboard to load or were possibly loaded outboard. Battles were not broadside duels, but individual aimed shots fired by the gunners when each gun came to bear as the steersman brought the ship around. Our guns are used to fire blanks as salutes. There are 10 gunports, 4 on each side, 2 in the stern.

Original smallarms included matchlock muskets, cudgels, pikes, halberds, swords and knives. Officers may have had technically superior snaphaunce, wheellock or doglock pistols which were all current at this time.

VOC ships were armed merchantmen acting with government authority. They were, in effect, an extension of the government, and acted in war

as navy vessels, with the crews as fighters (whether they signed on to fight or not). In addition to naval battles and combat against Indonesian warriors in *kora-kora* (war vessels), fighting included shore attacks and hand-to-hand combat.

Life Onboard

Most aspects of shipboard life were controlled by the authority of the captain. Janszoon was well-respected by his crew and was held in high esteem by his employers. He is recorded as being one of the few captains who never suffered disobedience or mutiny from his crew, but if necessary he would have had recourse to the Letter of Articles. Introduced in June 1595, there were 42 articles governing most aspects of daily life at sea with severe punishments for breaches of these articles.

The original *Duyfken* had a crew of 20 but could have carried passengers at times, and probably did so.

The names of a few persons who sailed on *Duyfken* are known.

1601–1603 voyage; Captain and Upper Merchant Willem Corneliszoon Schouten, Upper Merchants Jacob de Feyter, Pieter de Weel and Claes Gaeff, Upper Steers-man Abraham Francken and rescued castaway Francois, discovered in Mauritius.

1603–1604 voyage; Captain Willem Janszoon.

1605–1606 voyage; Captain Willem Janszoon, Upper Merchant Jan Lodewijk Roossengin.

1607–1608; *Duyfken*'s last commander was Cornelis Laurenszoon. *Duyfken* replica's crew numbers 18–20.

When *Duyfken* loaded to undertake her 1603–4 voy-age to the East Indies, she is known to have carried 7 sacks each containing 200 "Ryalls-of-eight" ("pieces of eight"), copperwork worth f 179-14-0, ships tools worth f 2-12-0, 12 French "pipes", (wine casks of 630 litres), 33 tuns and 2 demi-tuns of beer (tun=150 litres).

Other provisions that she is likely to have carried included live piglets, goats, poultry, flour, ships biscuits, bread in 74 lb barrels (33.63 kgs), butter, cheeses in barrels, dried, smoked and salted meat and fish, grains, meal, legumes. Water, fruit, poultry, eggs, oranges, coconuts, fresh fish etc., were replenished as opportunity arose.

Meals were prepared on deck. Cooking was done in a wooden, metal-lined box, or metal box, open at the front and top. Sand was spread in the bottom and a metal rod to suspend pots and food for cooking was

run from side to side. No-one was permitted to cajole the cook or appear at the galley seeking extra food. Meals included fresh caught fish, also fowl, and salted meats, probably roasted, boiled or stewed. Porridge and gruel were popular. When fresh food was exhausted or unobtainable, the crew ate only salted, smoked and dried supplies. The crew had to be content with what was served and were punished for wasting or refusing food or drink. Wine and beer were consumed in preference to water, as water carried in barrels soon went foul. Water was mixed with wine and beer to extend supplies of both. The issue of rations was strictly controlled.

Health problems such as scurvy and “bloody flux” (severe dysentery) were reduced by attention to diet and hygiene, but on long voyages scurvy would appear after about 6 weeks on preserved rations, unless fresh fruit or vegetables were acquired. The Dutch and other European mariners knew of the benefits of citrus fruits as a preventative/curative for scurvy, but the death toll was often high. In the Indies the diet included rice, poultry, melons, bananas, sago, eggs, onions, garlic, water and arak. Hygiene was addressed by regular cleansing of decks and quarters by the morning watches. No man was permitted to wash, perform bodily functions or have damp clothing below decks. The constable doubled as barber-surgeon for treating wounds and disease, and sail-ors were given great care and consideration during illness. Our crew have the benefits of modern hygiene, medicine, nutrition, and food preservation techniques.

Sleeping was on the decks on palliasses, but it is possible that dismountable wooden framed bunks were used, as these have been described for larger vessels in *Duyfken*’s fleet. These bunks had to be dismounted to allow room to work the guns during battle. (Hammocks were known at the time but there is no evidence of their use on Dutch ships). Our crew sleep on the deck, but using modern palliasses, or in hammocks.

Time was kept by sandglass and bell (or gong, drum). There were probably three watches on *Duyfken*, and sleeping on watch was strictly prohibited.

Navigation was by sun or star shots to find latitude, using cross-staff, backstaff, or astrolabe. It was generally not possible to calculate longitude. Instead, various techniques were used, including “running down the latitude”, to reach the desired destination. The use of sun fixes, compass to maintain course, and sandglass and chip to record speed, generally allowed the navigator to find a destination. Officers of the watch recorded the ship’s speed and compass headings into the log from the traverse board. The captain kept an overall record of events in the ship’s journal. *Duyfken*’s journals are lost, but we know something of the contents of the journal of the voyage to Australia because other 17th

century captains sailing to northern Australia make reference to it in their own journals. Our ship has modern GPS navigation.

Social barriers segregated the officers and men. The Master was accommodated aft in the captain's cabin and the Upper Merchant with him. There is evidence that officers received a greater allowance of liquor than the men and this may have been extended to certain other provisions. The fear of mutiny resulted in strict rules against entry to the powder magazine and the unauthorized firing of muskets or cannon. Ordinary sailors were not permitted to bring on board weapons amongst their personal gear in sacks or chests. Our modern captain and crew have similar quarters to the original crew, but share all aspects of life on board.

Glossary of Facts and Figures

Amsterdam Foot: 11 Amsterdam thumbs (*duim*) (25.6mm). These *duim* correspond closely to English inches (25.4mm), thus, an Amsterdam foot (281.6mm), is about 11 English inches (279.4mm). The draft markings on the bow and stern are in Amsterdam feet.

Anchors: *Duyfken* has two replica anchors as well as modern types.

Arak: spirits purchased in the Indies.

Astrolabe: a device for measuring the angle of the sun above the horizon.

Backstaff: a device for measuring the angle of the sun above the horizon.

Ballast: heavy material stowed low in the ship to give stability. Most VOC ships were ballasted with small bricks which were used for building in the Indies. Our *Duyfken* will be ballasted with bricks and some fixed ballast.

Banda: a small group of islands in the Banda Sea were amongst the most important of the Spice Islands producing all the World's nutmeg and mace. From Banda, *Duyfken* set out on her 1605–06 voyage to Australia.

BCE: Before Common Era, (equals BC).

Beam: width of vessel (6.01m), also deck beams:-bear-ers for deck planking.

Belaying pins: removable wooden pins to which ropes are belayed (tied off).

Binnacle: wooden box containing compass and other navigation aids.

Blocks: pulleys made from a wooden block containing a sheave (pulley wheel). Used in the running rigging of a ship. Similar devices with no moving parts are **deadeyes** and **euphroes**. Around 155 hand-made hardwood blocks, deadeyes and euphroes are used in the standing and running rigging.

Ship's boat: oak, flat bottomed, clinker built, about 5m overall, equipped with a small windlass. Usually stowed on the gundeck at sea, towed astern during battles. Used to collect provisions and water from shore, exploration and landing, tasks such as moving anchors, investigating reef hazards, etc.

Botte Loef: (pro. botta luff) a notched boom for ex-tending the tack of the foresail.

Braces: lines to trim the yards and angle of the sails.

Break: the step between decks.

Caulking: Dutch ships of Duyfken's time had animal hair and tar, or sphagnum moss between the plank seams. The replica is sealed with a modern product.

CE: Common Era, (AD).

Chain plates: iron plates located below the channels on the sides of the hull. They secure the deadeyes which connect the shrouds to the hull.

Channels: "chain wales": in *Duyfken's* day these were usually a thickening of the **wale** at the point where it was needed to hold the shrouds and deadeyes out from the hull. Much wider, separate, timbers were used later.

Compass: the compass of 1600 had 32 points marked on the rose. Compasses today have 32 points and/or 360 degrees. One point equals 11.25°

Constable: usually a senior man responsible for controlling aspects of shipboard life. He controlled access to, and issue from, stores, magazine and galley. He was chief gunner and also acted as barber-surgeon.

Crew: originally 20, but she was sailed with less when battle or disease depleted the numbers. Provisions lists show complement could have been as high as 33 in Nov. 1603.
The modern crew number approximately 20.

Cross staff: a device for measuring the angle of the sun above the horizon.

Crosstrees: timbers fixed across the masts to spread the topmast rigging. Part of the “top” (platform).

Crowsnest: *Duyfken*’s tops were not “crowsnests” — they are simple platforms formed by the **crosstrees** and **trestletrees**.

Deadeyes: triangular or pear-shaped blocks of wood without sheaves, usually pierced with three holes and used to set up the **shrouds**.

Decks: five levels of deck; main or gundeck, after peak, quarter deck, forecastle deck, and poop.

Decorations: the carving and painting of a ship. Our carver is Jenny Scrayen.

Discipline: covered by Letter of Articles introduced in 1595, which had 42 articles covering most aspects of shipboard life.

Duyfken: (pro: Doh-f-kin) “Little Dove”, from the bible, Noah’s messenger.

Euphroes: (pro. yewfrows) wooden blocks of wood pierced with a number of holes to separate multiple ropes such as **martnets**.

Fathom: Depth measurement, 6 Amsterdam feet (1.69m).

Figurehead: It is not certain that the original *Duyfken* had a figurehead. VOC ships of this time were usu-ally fairly plain but, if present, figureheads were usu-ally a red-painted lion.

Fire bending: the bending of oak planks is achieved by placing the plank in a frame with a fire underneath.
The plank is heated until heat makes the timber soft and plastic, allowing the plank to be gradually bent or twisted by means of weights and levers. Cooling allows the timber to harden again, and the plank takes on a permanent set.

Floors: the floors are transverse frame timbers in the bilge of the ship which cross the keel.

Futtocks: originally “foothooks”, all frame timbers that are not **floors** are called futtocks, 1st futtock, 2nd fut-tock *et seq*. The 1st futtocks sit between the floors at the curve of the bilge, these are called “sitters” by the Dutch. 2nd futtocks continue up the side of the ship, etc.

Galley stove: metal box, open at the front and top with a layer of sand in the bottom. Standing on the deck. Food could be roasted or boiled in pots suspended from a horizontal bar: stews, gruels, boiled or roasted meats and vegetables. Whenever possible, preserved provisions were supplemented with fish, seabirds and fresh fruits, vegetables and meats, etc.

Gelderlandt: (pro. Khelderlunt), province of the Neth-erlands, name of Admiral Harmenszoon's flagship. The three "from life" drawings of *Duyfken* were found in the journal of the *Gelderlandt*.

Gratings: hatches (openings in the deck) had gratings over them so crew would not fall through them and at the same time allow ventilation. These gratings would be covered with a tarpaulin in foul weather.

Great South Land: (*Terra Australis Incognita*) the theo-retical land-mass, long thought to exist in the south-ern hemisphere to "balance" the land-masses of the north. *Duyfken's* voyage of discovery in 1605–06 looking for Nova Guinea, a part of the *Terra Australis Incognita*, resulted in the first-known European contact with Australia. This discovery and charting of Australian coast was the first of many by the Dutch, resulting in about 60% of the coastal outline being known in the early 18th century. The voyage of Cook, 164 years after *Duyfken*, when he claimed the eastern part of the continent for the British Crown, saw most of the remaining east coast charted.

Glasses: sandglasses for keeping watch times, prob-ably half-hour and four-hour glasses.

Halliards: ropes for raising sails, yards, etc.

Hatches: openings in decks to permit access in and out of the hold or below decks.

Heads: the ship's toilets or "heads", located in the beakhead.

Hemp: plant fibre (*Cannabis sativa*) used in the manu-facture of rope, sails, oakum, etc.

Hold: Cargo and general goods storage area below the lower deck. *Duyfken* could carry 25–30 lasten (50– 60 tons).

Hutch: small steersman's compartment under the break of the poop, containing the binnacle and whipstaff.

Hygiene: strictly enforced by Letter of Articles. Below decks area cleaned and main deck washed daily by morning watches. Personal ablutions and toilet, damp bedding and clothes prohibited below decks.

Indies: West Indies, the islands of the Caribbean Sea. East Indies, the islands of the Indonesian archipelago.

Jacht: (pro. yukht) from the Germanic word “jagd” = “hunt, pursue”. *Duyfken* was purchased as a jacht whose function was to scout ahead of the fleet and to pursue and capture enemy vessels.

Janszoon: (pro. Yuns-zonn) Willem Janszoon, skipper of the *Duyfken* was Australia’s Christopher Colum-bus. First mentioned in historical records 1598, died 1638? Parents Volckert Janszoon and Elizabeth Claessens of Volckert. It is not known whether Willem married.

Java: Ancient centre of power and culture in Southeast Asia. Batavia, built beside the town of Jakarta in West Java, was the centre of administration of the Dutch East Indies and the spice trade. Jakarta is now the capital of the Republic of Indonesia.

Journals: officers of the watch kept a detailed log of daily events and progress. The captain was required to maintain a journal which incorporated much of this log detail and provided a record of explorations, dis-coveries, trade developments, etc. The ship’s journal remained the property of the VOC and was used by them to provide valuable information to other cap-tains and expeditions. It is likely that other officers also kept their own diaries.

Knees: angled timber supports called knees are used in the ship, there will be about seventy.

Knights: sheaved posts, often with carved knights’ heads on top, (hence the name), rising through the deck near the main and fore masts. Used to reeve and tie-off the **halliards**.

Lanyards: short ropes passing through deadeyes for tightening shrouds. General name for short pieces of rigging with no specific name: sailors carried knives secured around their neck or belt by a cord lanyard.

Last, Lasten: (pro. lust, lusten) Dutch measure of weight. 1 last = around 2 tonnes, 2 lasten = 4 tonnes, etc.

Lateen sail: triangular fore and aft sail on the mizzen mast.

Lighting: candles and lanterns. Fire was generally pro-hibited in the hold except under supervision of an of-ficer.

Lead: for sounding depth of water. Concave on bottom to allow “arming” with sticky tallow to pick up sea-bottom material for identification and noting on charts. The rope is knotted at intervals about the measure of a man’s outstretched arms (English fathoms, 6 feet).

Letter of Articles: 42 articles governing most aspects of shipboard life and behaviour. Introduced in 1595.

Liquor: gin, brandy, arak, wine, beer, in large quantities (Nov. 1603 took on board 33 tuns and 2 demi-tuns of beer, tun = 150 litres = 4950 litres. Also 12 French pipes of wine, pipe = 630 litres = 7560 litres).

Log: a ship's daily occurrence book, (see **journal**). A means by which the speed of the ship can be calculated (logged). By noting the time taken for a chip of wood thrown over the side, (or a remarkable patch of froth) to pass between two marks cut 40 feet apart on the ship's rail, the speed can be determined. If the chip takes 12 half seconds to pass the second mark, 12 is divided into 48, giving 4; this represents 4 geographical miles per hour, or 4 knots, or 1 Dutch mile per hour. (Taken from an 18th century description of how the Dutch estimated speed).

Martnets: a series of lines attached to the leach (side edges) of the sail. When lowering the yard in preparation to furling sail, the marnets attached to each leach take up the sail and assist the furling.

Masts (bow to stern): fore, main, mizzen. Masts and bowsprit are of pine, (*Pinus pinaster*), of pole type, scarf joined and wooded for strength.

Moluccas: group of islands in the Indonesian archipelago also known as the Spice Islands and now called Maluku.

Moot: hand-turned device for turning dowels or **trunnels**.

Myle, Mylen: (pro mail, mailen) Dutch/German measure of distance. 1 myle = 4 nautical miles, 2 mylen = 8 nautical miles, etc.

Navigation: finding latitude by taking a sun shot at noon using a **backstaff**, **cross-staff** or **astrolabe**. With the midday angle of the sun above the horizon and a set of tables for seasonal adjustment the latitude could be calculated. Longitude could not be determined so easily then, and "running down the latitude" was the accepted practise. This meant sailing south or north, to the latitude of the target landfall and then running east or west along the latitude until you found it. Course on the latitude was maintained by compass heading and checked with daily sun fixes. Depth soundings might be taken hourly in shallower waters approaching land. Navigation was usually adequate but errors sometimes led to disaster.

Oak: (*Quercus robur*) hardwood imported from Latvia, a Baltic State, for building *Duyfken*.

Oakum: mixture of tar and unlaidd hemp rope used for caulking seams. In *Duyfken's* day sphagnum moss was sometimes used. Today oakum is used but the seams are paid (sealed) with a modern compound in-stead of pitch.

Pumps, bilge: two elm-tree pumps (hollow wooden tube reciprocating pumps), located directly forward of the steersman position.

Pump dale: the gutter which directs the bilge pump discharge to the scupper.

Ratlines: ropes across **shrouds** for climbing into rig-ging.

Registration: *Duyfken* is registered in Australia and home-ported in Fremantle.

Rigging: the “standing” rigging comprises the stays and shrouds that hold the masts in place. The “running” rigging is used to hoist and control the sails.

Roossengin: (pro. Ross-in-kheyn), Jan Lodewijk Roossengin was Upper Merchant on the voyage to Australia. Variously spelt: Jan Lodewycksz (= Lodewyckszoon) van Roossengin, Senor Jan Rossengin, Jan Rossingen, Jan Lodewijs van Rosingyn, J. Rosingeyn, Jan Rosingein. Roozengein is the name of one of the Banda islands and is not a normal Dutch name.

Ropes: most standing and running rigging ropes are traditional hemp ropes. Sizes include 60mm, 48mm, 32mm, 24mm, 20mm, 16mm, 12mm. About 3.5 kilometres of rope are needed to sail *Duyfken*.

Rudder: made of oak, approx. 4.5 metres high and about 0.35 tonnes in weight.

Sails: Square sails: Bowsprit = spritsail; Foremast = fore course, fore topsail; Mainmast = main course, main topsail. Triangular sail, fore and aft rigged; Miz-zenmast = lateen.

Sailing speed: the original *Duyfken* is known to have achieved 6–7 knots and was probably capable of faster turns of speed. Our *Duyfken* can make 8 knots, and might record speeds above this at times.

Scurvy: a disease, caused by lack of vitamin C. Ap-pears after about 6–8 weeks with no Vitamin C as swollen gums, lethargy, fever and eventually death. The Dutch were well-aware of citrus fruit and fresh vegetables as a cure and preventative for scurvy.

Scupper: drain hole in side of ship at edge of deck, often fitted with a flap or valve to prevent the ingress of water.

Sheave: pulley wheel in block.

Sheets: ropes used to trim or flatten sails — not the sails themselves.

Shrouds: heavy rope standing rigging. Set up from mast to the side of the hull, shrouds give lateral support to masts.

Shipwrights: Master shipwright was Bill Leonard. He led a team of shipwrights and apprentices building the *Duyfken* replica.

Sounding: measuring the depth of the sea. see **Lead**.

Spice: the general name for the leaves, bark, seeds, etc. of certain plants, several of which only grew in the Spice Islands. Because of the enormous value of Spices sold in Europe, several nations vied to control and monopolise the trade. The Dutch emerged successful. Spices such as cloves, nutmeg, cinnamon, pepper and mace were important for food flavouring, emblems of wealth, and medicines to the people of Europe.

Stays: heavy rope standing rigging giving fore and aft support to masts.

Supercargo: an abbreviation of the term Superintendant of Cargo, a representative of the company charged to look after their ship, goods and trade interests during the voyage. Also called merchant or upper-merchant.

Swivel guns: small anti-personnel, sometimes breech loading, guns mounted in a traversing yoke or swivel on the rail. Fired metal or stone shot to sweep as wide an area as possible and repel boarders or mutineers.

Ternate: (pro. Tear-na-tay) important spice island on the western side of Halmahera. It was here that *Duyfken* finally succumbed to hard usage and possibly battle damage, and was condemned.

Texel: (pro. Tayssel) the westernmost of the Wadden Islands chain along the coast of the Netherlands. From this roadstead, port and pilot station many Dutch VOC ships sailed for the Indies.

Tiller: the wooden beam protruding from the top of the rudder into the hull. Connected to the **whip staff** which allows the steersman to control the ship from the deck above.

Timber: the ship timbers are of oak (*Quercus robur*), masts and spars of pine (*Pinus pinaster*).

Trestle trees: timbers attached to the mast and aligned fore and aft, with the crosstrees they form a platform and support the crowsnest on a larger ship.

Trunnels: treenails are large dowels of wood (with end wedges), used for fastening planking to the frames and deck planking to the beams.

V.O.C: *Verenigde Oost-Indische Compagnie*, (pro. Verain-ukhde Oost-Indisher Companay) the United East Indies Company: *Duyfken's* owners.

Voyages: c.1595–1600 uncertain, possibly engaged in coastal trade, privateering or smuggling.
1601–1603 with Moluccan Fleet of Admiral Harmenszoon to the Spice Islands and return.
1603–1604 with VOC fleet of Admiral van der Haghen to Spice Islands.
1605–1606 inter-island duties and voyage of exploration to Australia.
1606–1608 inter-island trading and fighting, declared unrepairable and condemned in 1608 at Ternate.

Watch: spell of duty, usually four hours. Also the team of crew members on duty on each watch.

Windlass: horizontal wooden winch for raising anchors.

Whipstaff: vertical pole connected through the deck to the tiller by which the steersman steers the ship.

Interesting Bits and Pieces

Beakhead The structure sweeping upwards from *Duyfken's* bow is the beakhead. Although the beakhead was primarily a decoration it was soon recognised as the ideal spot to put the ship's toilet — it is built out over the water. Although ships no longer have a beakhead the toilets on ships are still called the “head” or “heads”.

Cannabis Cannabis or hemp was used for making ropes and sails. Canvass is another spelling of can-nabis and is still used to mean strong heavy cloth. *Duyfken's* rigging is hemp rope and her sails are made of flax which is very similar to hemp.

Forecastle The built up area in the bow is called the forecastle, usually pronounced “Foke-sel”. Originally the fore castle was a castle from which to defend the ship or attack enemy ships. By *Duyfken's* time the aftercastle or “summercastle” at the aft end of the ship had become much higher and taken over the forecastle's original role. Later in history the fore-castle became the accommodation for the sailors (but not the officers).

Hammocks Hammocks were invented by people of South America. By *Duyfken's* time European visitors to South America had seen hammocks and described them, but their advantages for use on ships had not been seen. *Duyfken's* crew probably put their straw mattresses wherever they could, looking for the least damp spot available. The officers had their cabin in the stern of the ship and probably slept in cots.

Jacht *Duyfken* was a "jacht" (pronounced yacht with a hard "ch" as in "loch"), a name which we now associate with boats used as pleasure craft. But the name jacht is from "jachen" meaning "to hunt, pursue, or hurry", and jachten (plural of jacht) were fast armed ships, often used for piracy, privateering and smuggling. When sailing became a sport, small jachten were used as "spiel jachten" or "play-yachts".

Oak Like the original ship, the *Duyfken* replica's hull is built from oak timber brought from the Baltic region. Oak was the ship builder's favorite timber in Europe; it is a tough, durable, easily worked, and not too heavy. And it is a beautiful honey colour. The *Duyfken* replica oak came from Latvia. *Duyfken's* decks and masts are pine which is lighter and available in long straight lengths. The maritime pine was grown in Western Australia and was provided by the Department of Conservation and Land Management, Western Australia (CALM).

Phantom The Phantom lives on the island of Banda Neira. Banda Neira is one of the tiny Banda group of islands in Eastern Indonesia. In *Duyfken's* time the only nutmegs in the entire world were grown at Banda. It was from Banda that *Duyfken* set out on her historic voyage to Australia. The Phantom calls the native people pygmies but they are of normal stature.

Poop deck The poop deck is the highest deck of the ship and the furthest aft. The highest part of *Duyfken's* stern is nearly six metres above the water-line. The name has nothing to do with poo or toilets.

Rig *Duyfken* is rigged with three masts and a bow-sprit projecting forward from the bow. A small square sail is set on the bowsprit, two square sails on the foremast and two on the mainmast. On the mizzen mast there is a single lateen sail, triangular in shape. Square sails are not necessarily square in shape but they can be set "square" across the ship, at right angles to the keel.

Skipper From the Dutch *Scheep* meaning "ship" and *Scheeper*. *Duyfken's* skipper on her first voyage to the Indies was Willem Schouten who later discovered and named Cape Horn after the Dutch port of Hoorn. The skipper on the historic voyage to Australia was Willem Janszoon (William Johnson is the English equivalent). Archived documents describe him as a sober and honest man who was well-respected by his crews.

*“Water, water everywhere,
But it’s not the stuff to drink.”* Water stored in wooden barrels soon became putrid. Sailors, and people on land normally drank beer. Tea, coffee and cocoa had not been introduced to Europe and only hermits and beggars drank water. On the long voyages to the East Indies it was difficult to carry enough beer. The log book of one voyage records after five months at sea: *“The second day of August we dranke our last Beere, and we began our first allowance to drinke water, four mutskins or measures everie day, and three of wine.”* In the Indies, the ships were able to replenish their supplies with distilled rice wine and palm wine called *Arak* which the sailors greatly appreciated.

Whipstaff The ship is steered with a whipstaff. The whipstaff is a lever that goes through a wooden bearing called the “rowle” and moves the tiller. The tiller is connected directly to the rudder. The helmsperson puts the whipstaff to starboard (right looking forward) the bottom of the whipstaff goes the other way — to port — the rudder blade goes to starboard and, hope-fully, the ship turns to starboard. Simple as reversing a road train. The person steering can see some of the sails and sometimes uses the set of the sails to steer the ship by — “steering by the wind” — but he or she cannot see ahead. In the open sea they would steer by the compass and when maneuvering the pilot or the skipper would give directions.

Who “discovered” Australia? Archaeologists believe that ancestors of the Aboriginal peoples first crossed the seaways between island Southeast Asia and the Australian continent about 60,000 years ago. Although New Guinea has been joined to Australia by a land bridge at times there has never been a land bridge to Southeast Asia. Dingoes first came to Australia about 4 – 5,000 years ago. Dingoes do not build boats or navigate, so some-one must have brought them, almost certainly seafarers from Southeast Asia. It has been suggested that people from China, Japan, Ancient Egypt, France, Portugal, Spain and other places came to Australia before the Dutch reached Australia — but there is no firm evidence for any of those claims. In 1606 *Duyfken*, captained by Willem Janszoon, reached the Cape York Peninsula, Far North Queensland, and an accurate chart of the Cape York coast was made. That is the start of Australia’s re-corded history, but the Great Southern Land had been discovered many thousands of years before.

Did sailors believe in a flat earth?

It is often claimed that sailors in the Age of Discovery were frightened by the idea that they might sail off the edge of the world. This is probably not right. The Romans, Greeks and Egyptians had discovered and accepted that the World is spherical.

Sir John Mandeville’s *Travels* (a 15th century, or possibly earlier, travel-guide for pilgrims to the Holy Land) contains the following Chapter Heading:

Chapter xx
of the evil customs used in the Isle of Lamary. And how the earth
and the sea be of round form and shape, by proof of the star that is
clept antarctic, that is fixed in the south

Mandeville was a famously poor geographer but he knew that the World is round.

Contrary to popular belief, the Roman Catholic Church never really opposed the idea of a spherical earth and officially accepted it with no fuss about 800 AD.

Galileo did not get into trouble for saying that the world is round. It was for saying that the earth goes round the sun — that the earth is **not** the centre of the universe.

Copernicus was responsible for that discovery but his publisher diplomatically said that it was easier to predict the position of heavenly bodies if you calculated **as if** the earth went around the sun. He did not say that it was actually the case.

Why was Australia not settled by Europeans until 1788?

There is an idea that discovery by Europeans inevitably led to settlement by Europeans. In the case of the Americas this is fairly true, although for three centuries after Columbus nearly all settlement was confined to the coast and most Europeans living on the continent were engaged in exploitation of resources, such as animal furs, for ex-port back to Europe.

A combination of fire-arms, fierce dogs, horses, the wheel, and either *chutzpah* or zealotry gave a relatively small number of conquistadors the ability to conquer the nations of Latin America very quickly. In the Old World things were different.

In Asia, major conquest on land was not possible. The Europeans were only superior in the design of their war-ships. The heavily armed galleon was superior in fire-power to any Asiatic vessel of the time. But the Portuguese, and other European powers that came later, could only get a foothold in Asia by threatening to disrupt the maritime trade of the region.

Whether the Europeans wanted conquest on land is another question. Control of the trade in valuable commodities was what they were mainly after. By the time Australia was discovered to Europe, the strategy of enforcing trade monopoly was well established. The Dutch and other Europeans were not looking for lands to settle. Settlement of North America took place because tobacco growing was profitable and because coastal settlements were needed to exploit the fur trade from

the interior. America was also used as a place to exile criminals. Had a valuable commodity been discovered in Australia, it is probable that some kind of settlement would have been established, but not with the 19th century imperative of occupying the land.

When the American War of Independence deprived the British Government of their American penal colonies they decided to start one in Australia. Later colonies were established because of a perceived threat that the French would claim the land if the British did not. Some of the early settlements were military garrisons with a small contingent of convict workers. It was not until the gold rush that large numbers of free settlers boosted the population and spread inland.

The Dutch East India Company

It was in 1595 that Dutch merchants first set out to sail to the Indies and break the Portuguese monopoly in the spice trade. The first fleet was led by Cornelis de Houtman who had previously sailed to the Indies with the Portuguese. The fleet was organized by the Compagnie van de Verre ("The Far-lands Company"). Financial it was not a great success, but showed that it could be done. A second fleet set out in 1598 and made spectacular profits on the spice cargoes they brought back to the Netherlands. Inspired by that success, many companies and fleets were formed and in 1601 more than sixty ships set out for the Indies.

The problem was that the competition between the companies was forcing up the prices for spices in the Indies, and an over-supply of spices reaching the Netherlands pushed down the prices there. Moreover, The many competing companies were unlikely to succeed against the single, state-organised, Portuguese trading organization.

Thus it was that the Vereenigde Oost-Indische Compagnie (VOC) or United East-Indies Company was formed on 20th March 1602, amalgamating the existing companies into six chambers representing six major ports that sponsored voyages to the Orient.

The six chambers were: Amsterdam, Middelburg (province of Zeeland), Enkhuizen (province of North Holland), Hoorn (province of North Holland), Delft, (province of South Holland), and Rotterdam (province of Holland).

The VOC was managed by a board of seventeen representatives appointed from the six chambers according to each chamber's financial might. Amsterdam had eight representatives, Middelburg had four and the other four chambers had one each, plus one extra appointed jointly by those four chambers so that the VOC could not be dominated by Amsterdam..

The VOC was given exclusive power to legislate, prosecute and punish, establish colonies, and even to make war, everywhere east of the Cape of Good Hope. It is very unlikely that any monarchy would cede such great, sovereign power to a commercial organization, but the Netherlands was a republic with maritime trade the foundation of its economic strength.

The company was set up by national share subscription, each share cost 3000 guilders and the initial capital was 6,500,000 making it the largest company in the world; a position it maintained for more than a century.

The VOC logo, probably the first example of corporate “branding”, was adopted on 28th February, 1603.

At the end of the 18th century the VOC was bankrupt and the government decided that it owned VOC assets. The company was dissolved in 1799.

Spices and Herbs

Spices and herbs are parts of plants that are eaten, or included in food during cooking and not eaten, for reasons other than providing nutrition. Herbs are mainly leaves and flowers, while spices are generally seeds, fruits, bark, parts of flowers, buds, roots and other parts of plants, but there is no completely logical distinction between herbs and spices. Herb is from the Latin *herba* meaning grass, and spice from Latin *species* meaning appearance or type (kind, sort) and implying the kind of goods of which only the genuine article — the right species — will do the job.

Historically, herbs and spices have been as much associated with medicine as with culinary art. Knowledge of herbal medicine must be as old as conscious learning (perhaps pre-dating human beings) because some animals, including dogs, instinctively eat plants for medicinal effects.

Probably all systems of medical knowledge have experimented with plant-medicines and fitted the results into their explanatory systems — ideas such as the balance of the four bodily humours and the yin-yang balance.

In Medieval Europe, it was believed that the colour of food was important to health: in winter one should eat food flavoured with red and yellow spices while in summer green and white herbs are more suitable. Spices and herbs were, of course, also used for flavouring.

The international trade in spices dates back to time before the megaliths of Stonehenge were erected in England and the Pyramids in Egypt. Cloves, which could only have come from the Spice Islands of Eastern

Indonesia, have been found in a Mesopotamian archaeological site dated about 3,000 BCE.

Nearly 4000 years ago a young man named Joseph with a taste for loud clothes was sold to “a company of Ishmeelites [who] came from Gilead with their spicery and balm and myrrh, going to carry it down to Egypt.” Genesis 37:25.

The cuisine of the Roman Empire made use of spices from distant lands including India, Sri Lanka, China and the Spice Islands. After the fall of Rome, piracy and banditry made the transport of spices to Europe more difficult and costly. The rise of Islamic power stabilized the situation in the late 8th century.

Through the Middle Ages, relations between Christian Europe and the Islamic Middle East were seldom good (the Crusades were not exactly the high-point of European diplomacy) and the supply of spices to Europe was always tenuous. It was the Venetians who were most prosperous in negotiating trade from the east and the prodigious use of spices in Medieval and Renaissance cookery attests their success in supplying a great range of spices in large quantities. Medieval European cookery has been compared to the style that remains current in the Middle East in its abundant use of spices and bold mix of strong flavours, sweet fruits, vegetables and meats. The spices from the Indies which were of great value in *Duyfken's* time were Pepper, Cinnamon, Nutmeg, Mace, Cloves, and it was the last three that could only be sourced from the Spice Islands.

Names of Spices and Herbs

The following table summarizes the common and scientific names of most popular spices and herbs, and the part of the plant they come from.

Common Name	Scientific Name	Part of the plant
Allspice	<i>Pimenta dioca</i>	Berries
Anise	<i>Pimpinella anisum</i>	Seed
Annato	<i>Bixa orellana</i>	Seeds
Basil	<i>Ocimum basilicum</i>	Leaves
Bay	<i>Laurus nobilis</i>	Leaves
Caraway	<i>Carum carvi</i>	Seeds
Cardamon	<i>Elettaria cardamomum/Amamum c.</i>	Seeds
Celery	<i>Apium graveolens</i>	Leaves, stems and seeds
Cayenne	<i>Capsicum annum</i>	Podlike berries
Chia	<i>Salvia columbariae</i>	Seeds
Chile Pepper	<i>Genus Capsicum</i>	Berries
Cassia	<i>Cinnamomum cassia</i>	Bark
Chives	<i>Allium schoenoprasum</i>	Leaves
Chocolate	<i>Theobroma cacao</i>	Seeds
Cinnamon	<i>Cinnamomum zeylanicum</i>	Bark
Cloves	<i>Eugenia aromatica</i>	Flower buds
Coffee	<i>Coffea arabica</i>	Seeds
Coriander	<i>Coriandrum sativum</i>	Seeds, leaves, roots
Cumin	<i>Cuminum odorum</i>	Seeds
Dill	<i>Anethum graveolens</i>	Leaves, seeds
Fennel	<i>Foeniculum vulgare</i>	Seeds, leaves, roots
Fenugreek	<i>Trigonella foenum graeca</i>	Seeds
Garlic	<i>Allium sativum</i>	Bulb
Ginger	<i>Zingiber officinale</i>	Rhizomes (roots)
Horseradish	<i>Armoracia rusticana</i>	Roots
Juniper	<i>Juniperus communis</i>	Berries
Mace	<i>Myristica fragrans</i>	Seed coverings (arils)
Marjoram	<i>Origanum marjoram</i>	Leaves
Mint	<i>Mentha sp.</i>	Seeds
Nutmeg	<i>Myristica fragrans</i>	Peeled seeds
Onion	<i>Allium cepa</i>	Bulbs
Oregano	<i>Origanum vulgare</i>	Leaves
Paprika	<i>Capsicum annum</i>	Fruit pods
Parsley	<i>Petroselinum crispum</i>	Leaves
Pepper	<i>Piper nigrum</i>	Berries
Pimiento	<i>Capsicum annum</i>	Fruits
Poppy seed	<i>Papaver somniferum</i>	Seeds
Rosemary	<i>Rosmarinus officialis</i>	Leaves or flowers
Safflower	<i>Carthamus tinctorius</i>	Flowers
Saffron	<i>Crocus sativus</i>	Flowers stigmas
Sage	<i>Salvia sp.</i>	Leaves
Savory	<i>Satureja species</i>	Leaves
Sesame	<i>Sesamum indicum</i>	Seeds
Shallot	<i>Allium cepa</i>	Bulbs
Star Anise	<i>Illicium anisatum</i>	Unripe fruits
Tarragon	<i>Artemisia dracunculus</i>	Leaves
Thyme	<i>Thymus sp.</i>	Leaves
Turmeric	<i>Curcuma longa</i>	Rhizomes
Vanilla	<i>Vanilla planifolia</i>	Seed pods

Remarks on Urban Legend #27: Medievals used Spices and Sauces to Cover the Taste of Rotten Meat

The urban myth that medievals used flavoring ingredients to cover the taste of spoiled meat has proved astoundingly persistent. Competent historians who know little or nothing about the culinary record retail it in survey texts. Articles in popular sources blithely subscribe to it. It's nonsense, and it's demonstrable nonsense. Below, I present three modern observations, and then some facts about what the primary sources from the time have to say on the subjects of rotten meat and sauces.

Modern observation # 1:

It has often been observed that medievals were exposed to meats of variable quality, and frequently of lower quality than modern meats. That is true, but not necessarily relevant.

Variable quality is different from variable freshness. We know, for instance, that major advances in agriculture and breeding resulted in a tremendous improvement in the overall quality of British beef in the 17th or 18th century (I am unsure which) — which is certainly not the same as saying that no beef before that was fresh. In general, to say that sauces masked the variable quality of ingredients is not to say that they were used to hide rot.

Modern observation #2:

It is also widely assumed that medievals aged meat. I am not certain, in fact, that this is so; I can find no reference to it in the recipe corpus, and I find many references that clearly contradict it. It is a relatively common mod-ern practice, especially with game; but I am far from certain that it was a medieval one. But supposing it was, again, that fact is not relevant.

Unless someone screws up, aged meat is not rotten. Aging must be done carefully, or the meat will rot. But the difference is clear, and it is easy to tell when it has happened. Aging is not a strange and exotic practice that Americans at home never encounter. Every fine restaurant I have ever been to (not to mention many just-sort-of-decent ones) ages its steaks. Aging can be perfectly safe; more to the point, when it goes wrong, that fact can be safely detected. Even by medievals.

Modern observation #3:

Some people have suggested that the sauces are intended to preserve the meat, and pointed out, for instance, that black pepper in fact has preservative qualities.

Well, yes, in the sense that if you take fresh meat, and cure it with lots of pepper, that can help prevent spoilage. Some meats and fish were

salted, to exactly that end; and it works. But if you take already spoiled meat, cook it, pour a pepper sauce over it, and serve it, what you get is spoiled meat with pepper on it. The same goes for salt, of course. Many, many recipes end by saying to pour the sauce over the meat and serve it forth. Whatever is going on here, it isn't preservation.

In the case of salt, it also isn't concealment, and it isn't a habit tending to hypertension either. Medievalists almost never left the salt in salt-preserved foods. They soaked them for a long time, in several changes of water, some-times boiling, to leech out the salt before preparing. I've seen half a dozen recipes for what to do if the meat is still too salty after normal treatment. None involve disguising the salt flavor; all involve getting the salt out of the dish. It makes for both a better and a more healthy dish — as they knew as well as we do. The medievalists understood that excess of almost anything can be bad. Concerning rotten meat:

The culinary corpus addresses the question of bad meat from a number of angles, and its testimony is telling.

1. Recipe number 58 from *Diuersa Seruicia*, the second collection reproduced in *Curie on Inglysch*, describes what to do with a joint of venison that has just begun to show signs of going bad in spots. It starts by cutting away the bad spots. Next, you soak it in cold water, then bake it slowly in the hearth for three days and three nights. After that, you apply saltpeter to the area around where you cut out the bad bits, then soak overnight in rainwater. I am not a microbiologist (or for that matter a culinary hygienist); and I am modern enough to want to just throw the thing out and do without venison. But my impression, from reading the recipe and thinking about it, is that even by modern standards, this ought to just about turn the trick.

Recipe number 57 describes how to prevent venison from turning (essentially, how to dress and salt it). They are very clear about not exposing it to air before salting. There are similar recipes in other collections, including one that preserves venison by immersing it completely in honey, and then sealing the container. They all look, from a modern standpoint, like preservation techniques. The whole business, both of preventing rot and of dealing with a joint some spot of which has begun to turn, involves a lot of time and effort. No one would go to all that trouble if s/he thought that it were okay just to add more sauce. The clear presumption is that eating bad meat is a bad idea.

2. More recipes than I could count say things like "Take fair flesh of the forequarters and ribs", or "take fresh brawn". The clear implication is that one is to use good meat (and that fresh is good).

One might suppose that the reference to “fresh” suggests that “rotten” is a recognized serious alternative. But that does not seem to be the way such recipes are written. In the same manner, one sees instructions to take “good” herbs but never bad ones, “fair” water but never foul, and so on. I believe that the adjectives actually serve as re-minders to have care with the quality of one’s ingredients.

3. If you look at household accounts for upper class households, you will find that they go through a stagger-ing number of animals. On the whole, the delay between slaughter and consumption of four-legged domestic animals does not seem typically to have been long enough to lead to rot.

It is true that some beef and pork was salted, even in upper class households; but market and household records suggest that the extent to which herds were culled in the fall, leading to all meat in winter and spring being pre-served, is typically exaggerated, at least for the classes for which we have evidence that sauces — or spices of any sort — were frequently used.

4. Chickens and unsalted fish were normally killed immediately before cooking. The recipes frequently specify that; in the case of fish, they often detail how to kill it, and the method varies depending both on the fish and on the recipe.

5. The *Menagier* actually discusses the case in which chickens might be killed beforehand, and talks about how long they can be kept, either in an airless environment or on ice. I don’t know whether medieval Europeans ever had regular access to ice houses capable of keeping ice solid all year (or to ice that had been kept in such ice houses, whoever controlled them), but one is described in Pliny, so the technology was not beyond grasp. Keeping food on ice was surely an option in the winter, and may have been possible in the summer as well. By modern standards, the *Menagier* is if anything conservative.

Concerning sauces and spices in general:

1. They won’t cover the taste. I know, in the simplest way possible. I’ve been doing experimental medieval cookery for over a decade. In that time, I have twice accidentally used a piece of meat that looked okay, but had in fact started to go over (once beef, once pork). Both were prepared with sauce. In both cases, it took one bite to tell me the meat was bad.

One might conjecture that I am making the sauces blander than they were made in period. Three points tell against that. First, I am using relatively fresh spices. There’s a medieval health book somewhere that says that spices are more suited to the delicate constitutions of re-fined persons than herbs are, because they are more delicately flavored.

Think that through, and consider what it says about the condition of the spices they were using, and how strong recipes made with them are likely to be.

Second, and independently, even assuming that their spices were fresher than ours, household accounts do not support the hypothesis of extravagant use.

Third, the reason I could tell that the meat was spoiled was not that the sauce was delicate, but that spoiled meat has a very distinctive and pervasive taste, that is difficult to impossible to cover.

2. Medievals in fact had elaborate theories of medicine that some of them, at least, applied to the design of spicery and sauces for particular meats; you can look them up and read them. For a single example: Magninus Mediolanensis wrote both an extensive health tract with a chapter on sauces (the *Regimen Sanitatis*) and the *Opusculum de saporibus*, a work entirely dedicated to sauces. The work describes in detail the characteristics of the meats, poultry, and fish that make them wholesome (in terms of the theory of the humors) and how sauces are to be used to correct the main item to the appropriate levels of warmth and moistness (not in the modern sense, but in the sense of the humors). There is considerable dispute within the culinary history community on how important these theories were to working cooks; and the dominant view remains that they really didn't matter much to most cooks. But it remains significant that the sauces in Magninus Mediolanensis's work in fact adhere to the rules he gives; and that they match the sauces in the English and French repertoires quite closely, including the question of what meats to prepare how and what sauces to serve them with. The material from the documentary record, including (but not limited to) the health handbooks, is particularly revealing on this issue. The information contained in it has at least three consequences. First, it helps measure the balance of modern against period sauces by information more reliable than projections from purchases for meals and the like. Magninus Mediolanensis is particular about the balance of ingredients (often to the level of giving mathematical proportions). His proportions do not result in sauces capable of covering the taste of rot, even using fresh modern spices.

Second, it tells us what at least a few of them thought they were doing. We may grant that what people are doing, and what they think they are doing, may be two quite different things. But it is hard to imagine that a medieval cook might think he was balancing the humors, when in fact he was trying to cover a nasty taste.

Third, it makes it screamingly clear that the medievals were aware of health hazards of improperly chosen or improperly treated ingredients (their reasoning may have been different from ours, but that much

matches), and that they understood that fresh is good.

In Summary:

Especially in the 19th century, when evolutionary theories with teleological overtones were in vogue, and every-one tended to think that the world was going from worse to better, it was widely believed that medievals used spices to cover the taste of rotten food. The view was widespread enough, and pervasive enough, that one still finds it even in respectable historical sources that have not done their culinary homework. (Culinary history remains something of a select niche; most historians know next to nothing about it.) The fact is, that not only does the record not support the view, it directly contradicts it, over and over and over and over and over again, and in dozens of different ways. No respectable culinary historian would give the hypothesis the time of day; we know that it is false.

What you see in these pages are my opinions. Nobody pays me to have them, and I doubt that anybody else would want to claim them.

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Anti-Fouling And Protection From Toredos

"Protective coating of wooden ship's hulls has been a major concern ... since time immemorial." wrote Thijs Maarleveld, head of Underwater Archaeology in the Netherlands.

There are two distinct problems:

1 toredo and other marine borers can eat a ship's plank
leaving it in sinking condition in a few months,
particularly in the tropics 2 tendrils of weed and barnacles clinging to a
ship's
bottom dangerously reduce its speed and
manoeuvrability. A ship in this condition is said to be
"foul".

The solutions to these two problems are sometimes confused. Today they are combined in anti-fouling paints.

Toredos are bivalve molluscs, related to mussels and oysters. They enter a ship's planking as micro-plankton and grow to be as much as a metre long, using their shells to chew tunnels through the planking. Toredos exist mainly in tropical and sub-tropical seas; they are not a problem in most European waters though in southern European waters smaller borers known as "gribble" cause some damage to oak hulled ships. Some timbers such as teak have natural protection from toredo, but European oak makes a very fine meal for them.

In *Duyfken*'s time the standard protection from toredos for ships sailing to the tropics was to cover the ship's bottom with a sealing layer of animal hair felt and tar. This was then protected by a layer of thin sacrificial planking. That layer was nailed in place with many thousands of square-headed nails, aligned so that there was little timber exposed between the nail heads.

An anti-fouling layer was applied to this protective layer. For ships operating in the tropics the anti-fouling was what the Dutch called "*harpuis*": the English called a similar concoction "herthpich" and later "white stuff", distinguishing it from the tar-based coating called "black stuff" that was applied to ships serving in temperate northern waters. Archaeological evidence shows that in *Duyfken*'s time some Dutch ships working in northern waters had *harpuis* applied direct to their planking which was scored with knives or axes to key the *harpuis*. *Harpuis* contained brimstone or sulphur mixed in natural resins, and sometimes contained lime.

Other solutions were tried. The Dutch East-Indiaman *Mauritius*, wrecked off the Gabon coast, Africa, in 1602 was sheathed with lead, as Roman ships had been. In the late 18th century copper sheathing over tar and felt was used by the British Royal Navy and by the mid-19th century was in very general use. The tarred felt was the protection against toredos and the copper was the anti-fouling layer.

There were large Indian, Arab, Chinese and Indonesian ships operating on the Indian Ocean and in South-east Asia when *Duyfken* sailed to the Indies. They were all protected from toredos and anti-fouled with various types of lime coating.

Lime was made by firing sea shells and/or coral to produce quick-lime. This was then slaked to produce slaked or hydrated lime (the basic ingredient of old-fashioned mortar). In Indonesia this is then pounded and finely sieved. Then it is pounded again for several hours with a small amount of coconut oil or *Calophyllum* (oil-nut) oil until it becomes a smooth, hot and highly sticky putty. This is spread evenly over the whole underwater body of the ship using spatulas made from the inside of green papaya skins — the only thing it will not stick to. It hardens quickly. A layer of lime-wash is painted over the putty layer to make a loose, friable surface so that any fouling is easily scraped off.

Knowledge of the Southland before *Duyfken*

Duyfken made the first historically recorded voyage to Australia.

The first voyagers reached Australia several tens of thousand years earlier.

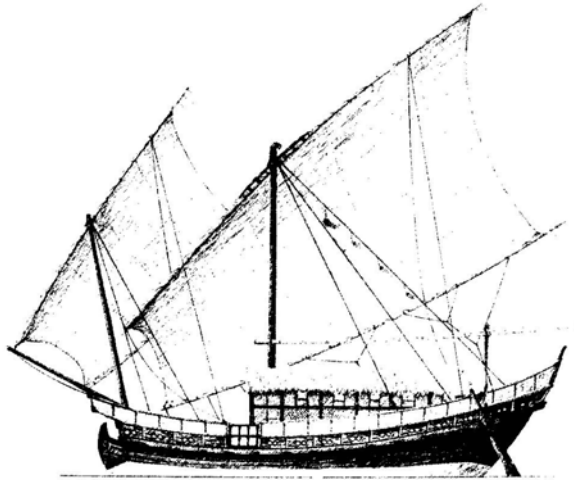
There is some evidence that Southeast Asian mariners knew of land to the south before the Dutch “discovery”. The record of that knowledge has become very confused. A misreading of Marco Polo gave rise to the idea that Java was either joined to a southern continent or separated from it by a narrow strait. Several names were ascribed to this southern land by geographers. These include Locach, Beach (pronounced Bay-ahk), and Laut Chidol (which is the Javanese for “South Sea” and is pronounced “Lout Keedol”)

An Italian traveller Ludovico da Varthema travelled in Southeast Asia some years before the first European ships reached the region. When travelling from Kalimantan to Java on a Javanese ship he asked the captain about various aspects of navigation and recorded clear and reasonable answers. He also asked about land to the south of Java. the captain said he had never been there himself but people did sometimes go there. In the winter it was cold and the day lasted only three or four hours. This makes good sense when one knows that the Javanese system has ten hours in a day (one Javanese hour = 144 of our minutes)..

A half Buginese, half Portuguese navigator named Eredia wrote about land to the south of Indonesia. Some of what he said is fantastic, some sounds like real knowledge. He said there was alluvial gold, it was dry and sandy and the indigenes were hostile and could only be traded with on the beach.

We know for certain from historical records that by the second half of the 17th century Buginese and Macassarese people from South Sulawesi were regularly voyaging to north Australia to collect and process sea food that they exported to China. They also traded with the Aboriginal peoples of the region. Some types of archaeological evidence and oral history suggest that regular voyaging to Australia by South Sulawesi peoples, perhaps the Bajau “Sea Gypsies” had started well before the 17th century.

The South Sulawesi mariners called north Australia “Tanah Marege” (Tana Maregh-ay with a hard “g”). They also voyaged to the Kimberley region which they called Kayu Jawa (Kye-you Jawa).



A Javanese ship (a "jong") of Duyfken's time.

Treatment of the Sick and Diseased: (Notes from the voyage of Jacob van Neck, 1598)

At the Isle of de Cerne, which the fleet named Mauritius, the sick and diseased from the ships were taken ashore and laid in tents and cabins until they recovered. They were probably suffering from scurvy and recovered quickly.

The sick were also taken ashore at Saint Helena on the return voyage. They were taken ashore on 9 December 1599 and brought back aboard on 31 December 1599.

Diseases mentioned are scurvy and bloody flux. People who died are described as being 'cast off the ship' unless they died while the ship was in port, in which case they were buried.

Notes From Daily Life In Rembrandt's Holland by Paul Zumthor

Peasants on land ate four meals a day **breakfast**: bread butter cheese milk beer **main meal** (Midday) soup and meat or soup, fish and meat ;salad or fruit dessert pancakes, waffles or rice pudding. Or cold meat and bread. **snack** (3 pm) bread cheese almonds or raisins hot or cold beer. **dinner** (8 or 9 pm) leftovers butter cheese porridge of stale bread soaked in milk.

Peasants ate vegetables such as peas, haricot beans. white cabbages, carrots, turnips and cucumber with dairy produce. Sailors lived largely on fish. gruel and cheese. Ordinary bread was made of rye-flour, barley, buck-wheat, oats, or even beans, and was black, sometimes soft and sticky, and always heavy. Potatoes were considered poisonous. Eggs were one of the cheapest sources of protein. Fish was available smoked, salted, or dried, as well as fresh.

Meat could be smoked or salted but was rarely eaten by the poorer members of society. Poultry and game were eaten by peasants and the nobility, but less so by the middle classes. Beer was the national everyday drink. Little water was drunk — it was often unsafe to drink. There were two main kinds of beer — simple and double. Double beer was strongly alcoholic.

Provisions List: 1600 fleet bound for the Indies

Bread (Ship's biscuits)
Spanish Wine
French Wine
Oil
Vinegar
Meat in brine
Brandy
Water
Garlic
Butter
Cheese
Various fish in brine
Beer
Groats (Cereal porridge with lard?)
Barley
Beans
Peas
Mustard Seed
Suet
Rice
Rape seed oil
Tabacco (early example of use — intended as anti-scorbutic)
Twenty barrels of scurvy grass with horse radish ("Mirredick")

Oak

Duyfken's hull is built of oak (scientific name *Quercus robur*) from Latvia in the Baltic region. The original *Duyfken* was certainly built of oak, and it is quite likely that the oak came from the Baltic region.

Oak is a hardwood and it is fairly hard. Balsa wood is a hardwood, but it is not hard. Huon pine is a softwood, but it is very hard.

Hardwood/softwood is a technical distinction between the timber from coniferous trees (softwood) and all other timbers. Conifers are the trees with their seeds in cones and with needles rather than broad leaves, all conifers except larch are "ever green".

Most non-conifers are deciduous — they lose their leaves at a certain time of year. The cellular structure of conifers is different from that of broad leaved trees — conifers are an older type of tree.

Oak is strong, reasonably easy to work, resistant to splitting, resistant to rot. It expands and contracts considerably with changes in moisture

content, but not as much, and not as quickly as some hardwoods such as jarrah.

Crime & Punishment Aboard a VOC East Indiaman.

Life on board a Dutch East Indiaman was a tough experience for the crew. Crime was dealt with severely and the punishments meted out were harsh and often extreme.

During their early ventures, the V.O.C. (Verenigde Oost-Indische Compagnie — United East India Company), issued instructions separately to each fleet and did not consider that it was necessary to incorporate them in a code of law to enforce discipline at sea. The Letter of Articles issued to one of the early voyages presents a picture of the discipline and discipline problems encountered on shipboard in those days.

All hands were admonished to be guided by God Almighty and to listen attentively to His word at prayer and sermon. Those who cursed or used His name in vain would be flogged at the mainmast and pay a fine to the poor. Whoever refused to obey orders from the skipper or his delegate received the death penalty. A court consisting of the highest officers had full jurisdiction, and the sentence was to be executed at once by the provost. Anyone who interfered with the provost or refused to lend assistance was punished as a mutineer. Desertion or the unauthorized use of a boat were punished by a severe flogging. No one was allowed to go ashore without permission or allowed to stay there overnight. Who-ever started a fist fight was thrown in irons and put on a diet of bread and water. If a shipmate tried to give the offender food or otherwise ease his lot, he received the same punishment. Pulling a knife, even as a threat only, was more serious. For a first offender, it meant three successive dunkings into the sea from the yardarm. For a second offense, he was to be keelhaunched. If a fight drew blood, the offending hand was nailed to the mast and the other tied behind the man's back. He would stay there until he was desperate enough to tear himself loose. In addition, he forfeited all his wages, one half going to the poor and the other into the common fund. If a man happened to kill his opponent, he was tied back to back with his victim and thrown overboard.

No one was to cajole the cook or even appear in the galley to try to get additional victuals. The crew had to be content with whatever was put before them, and those who were caught throwing food or drink overboard received corporal punishment. It was conceded that some drinking water or other liquid might be spilled upon the deck, but if the spot was greater than could be covered by a hand or a foot, it was not considered a permissible accident.

There were strict rules against sleeping on watch, entering the powder hold, and unauthorized firing of musket or cannon. To guard against fire, no light of any kind was allowed below decks without supervision by an officer.

“To prevent the stench from causing sickness the morning watches shall daily clean and wash the ships. No one shall defecate nor make water except at the head. Wet clothing may not be carried below decks, but first had to be dried.”

To protect the merchants, crew members were allowed to bring aboard chests and bales containing only personal gear. It was emphasized that all ships should re-main together and come to each other’s rescue in case of need.

There were 42 articles in all, and they covered just about every form of misconduct likely to be encountered.

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The Cradle of Colonialism. by George Masselman. 1963. Published by New Haven and London, Yale University Press.

Australia’s First Ship And A ScurvyLife At Sea - Nick Burningham

Early in the year 1606, during the stormy tropical wet season, the small Dutch jacht *Duyfken*, captained by Willem Janszoon (abbreviated Jansz.) made a landfall on the Cape York Peninsula of northern Queensland. Janszoon, who was commanding a voyage of exploration for the Dutch East India Company, surveyed some 300 km of the Cape York coast in remarkable detail and then surveyed Torres strait, some months before Torres got there.

Duyfken’s voyage is the first historically recorded voyage to Australia. In Fremantle, Western Australia, a replica of this highly significant ship has recently set sail for the first time and is now fitting out with circa 1600 furnishings and equipment to show visitors life on board a ship in the Indies spice trade.

Life on board was not comfortable.

Returning to Europe, valuable spices were crammed into all the available space. *Overijssel*, a contemporary jacht the same size as *Duyfken*, returned with her bread room full of spice cargo meaning that the provisions must have been up on the main deck taking up accommodation space.

Francois Pyrard was describing a Portuguese ship returning from Goa when he wrote:

"The [weather] deck of our ship was so laden with goods that they reached halfway up the mast. Outboard as well — on the railings and the ledges on either side you could see nothing but goods, provisions and bunks, that is little cabins where the sailors and other people lie down, covered with ox or cow-hides. In short the whole deck was so arranged that you could hardly move around on it ..."

Even allowing for some literary license and, perhaps, for things being worse on Portuguese ships, we can see that sailors were living as if they were camping on board. They were allowed to build their own shelters out on the weather deck. The idea of the crew living in tiny leather boxes like cubby houses made from cardboard boxes is extraordinary! We might construct an example on *Duyfken* but cluttering all the decks like the most desperate kind of refugee camp is probably taking authenticity too far.

Raleigh and others wrote condemning this building of these "kennels" that bred vermin and disease, but did not propose how the crew could be more satisfactorily accommodated.

Another practice that Raleigh condemned was building the galley (without a chimney) down in the bottom of the hold. Apart from the smoke, it filled the bilge with spilled food-stuff causing terrible stench and rotted the timbers of the ship. However, Dutch ships had their galley fire in a large box on deck.

These kinds of details are seldom recorded and rarely survive in the archaeological record. We learn that Dutch ships did not normally have their galleys in the hold because when arctic explorer Willem Barentsz.' ship was trapped in the ice at Nova Zembla in 1597 his journal records that the men coming up on deck to aid their ship-mates chased by a polar bear were blind from smoke because the galley fire had been moved down into the hold to counter the terrible cold.

Reading these journals to reconstruct life on board reveals many fascinating details, not least in the area of shipboard health, scurvy and diet.

On 9th December, 1599, two Dutch ships, *Gelderlandt* and *Zeelandia*, returning from the East Indies reached the island of Saint Helena in the South Atlantic and quickly dispatched a landing party:

". . . some of our people running up into the land to seeke after Cattell, brought some aboard our Shippes with them which made our mariners very gladde: but wee found no Orenge, whereof wee had most neede, for those that were trubled with the scurvie disease."

The following day they found a valley with orange trees and picked some 4000 fruit “which marvellous refreshed and cheered us all.”

Contrary to popular belief, the value of citrus as a cure for scurvy was understood by those 16th century mariners and the planting of orange trees on St Helena attests Portuguese knowledge. Almost as soon as they discovered the uninhabited island they stocked it with cattle and goats and planted citrus trees so that their ships on long voyages to and from the Indies could take provisions and refreshment there.

Other anti-scorbutics were widely known in the 16th century. The survivors of Willem Barentsz.’ last attempt to sail north around Russia to the Indies had spent a long autumn, winter, spring, and most of a summer, trapped by ice on the arctic island of Nova Zembla, living on ship’s stores, arctic fox and the occasional polar bear. By the time summer’s slight thaw let them escape in the ship’s boats they were suffering from scurvy and their first concern on reaching vegetation was to search for “leple leaves”.

“ . . . we found great store of Leple leaves, which served us exceeding well, and it seemed that God had purposely sent us thither : for that as soon as we had eaten them, we were presently eased and healed . . . we ate them by whole handfuls together, because in Holland wee had heard much spoken of their great force, and as then found it to be much more than we expected.”

The next day they collected more leple leaves:

“ . . . & still more recovered our healths, and in so short time, that we could not chose but wonder thereat, so that as then some of us could eate bisket againe, which not long before they could not do.”

Leple leaf, *lepelblad* in Dutch, is *Cochlearia*, prob-ably *Cochlearia officianalis*, sub species *groenlandica* on Nova Zembla, also known as spoonwort and scurvy grass. Clearly the explorers had been briefed in Holland about the anti-scorbutic they could expect to find in arc-tic regions.

Dr Woodall's “The Surgeon’s Mate” published in 1617 described scurvy and listed “Lemmons, Limes, Tama-rinds, [and] Oranges” as particularly efficacious cure. Woodall recommended the English East India Company provide lemon juice for their sailors. Although the chem-istry of vitamin C would not be unravelled for another three hundred years, Dr Woodall was essentially right.

So, how have we come to the idea that the problem of scurvy was only solved in James Cook’s time?

During the 17th and 18th centuries medical science developed in complexity enabling theorists to propose wonderfully unlikely experiments in a search for more arcane and unpleasant tasting cures. Joseph Banks, sailing with Cook on *Endeavour*, recorded on his journal for 16 June 1770 that Tupaia the Tahitian chief had “every symptom of inveterate scurvy notwithstanding acid, bark and every medicine our Surgeon would give him . . . “



Barentsz.' ship trapped in the ice at Nova Zembla

Tupaia had his own cure, as Banks noted just two days later:

“Tupia who had employd himself since we were here angling & had livd intirely on what he Caught was surprizingly recovered.”

In 1671 a long voyage provided Dr Lind, a Royal Navy surgeon, with a good sample of scurvy sufferers to ex-periment on. The following cures were tried, each on two men: cider, seawater, vinegar, elixir vitriol, a con-coction “of garlic, mustard seed, rad. raphan., balsam of Peru and gum myrrh”, and two lucky men were treated with oranges. Lind reported unequivocally that nothing had any effect except the oranges which worked power-fully and almost immediately.

Most self-sacrificing amongst the scurvy theorists was William Stark (1741?–1770). Born in England of Scot-tish parents he studied medicine at Edinburgh, London and Leyden. Returning to London in 1769, Stark began dietary studies on himself that culminated in his death after seven months. He began with a diet of just bread and water for thirty-one days, he then added other foods one at a time: olive oil, milk, roast goose, boiled beef, fat, figs and veal. He recorded that after two-months the gums of both jaws were red and swollen and bled when pressed, this was undoubtedly the onset of scurvy.

Dr Lind noted in his experiments that the men given oranges ate them “greedily”. Their bodies were telling them what they needed. The hunger for vitamin C is equally evident in Richard Dana’s “Two Years Before the Mast” when he describes how the scurvy afflicted men on the *Alert* ate raw onions acquired from a passing ship:

“And a glorious treat they were. The freshness and crispness of the raw onion, with the earthy taste, give it a great relish to one who has been a long time on salt provisions. We were perfectly ravenous after them. It was like a scent of blood to a hound. We ate them at every meal, by the dozen; and filled our pockets with them, to eat in our watch on deck . . . “

James Cook and his crew on *Endeavour* were keen on onions. For example: 17 September 1768 “Issued to the Whole Ships Company 20 Pounds of Onions per man” and only two days later “Issued to the Ships Compney 10 pounds of Onions pr Man”. Cook has of-ten been lauded as the discoverer of the cure for scurvy and sometimes discounted as the man who set back by twenty-five years the implementation of the known cure by recommending malt (no vitamin C) and sauerkraut (little vitamin C) as particularly efficacious anti-scorbutics. Although it is true that Cook made those mistaken recommendations, his practice during *Endeavour*’s voyage suggests greater insight or intuition. While on the coast of New Zealand Cook recorded on 27 October, 1769: “. . . the other place I landed . . . I got as much Sellery and Scurvy grass as loaded the boat.” The wild celery was *Apium prostratum* or *A. filiforme* and the New Zealand species identified as scurvy grass was probably *Lepidium filiforme*. The following day Cook explains that:

“Sellery . . . boiled with Portable Soup and Oatmeal every morning for the Peoples breakfast . . . I looked upon it to be very wholesome and a great Antiscorbutick.”

The cure for scurvy had long been obvious, the problem was acquiring and preserving adequate quantities of anti-scorbutics for very long voyages. James Lancaster, leading a fleet to the Indies in 1601, issued three spoonfuls of lemon juice to his men everyday un-til it ran out. But adequate quantities of lemon juice would have been hard to come by. And if the popular stereo-type is correct, the sailors would have been much more concerned to see adequate quantities of beer and wine go aboard at the start of a voyage.

In fact the beer and wine were important to their health. Water stored in wooden barrels quickly became “blacke as kennel [sewer] water” as the journal of *Gelderlandt* notes on 20th September, 1598. It could be used for cook-ing but it was only regarded as a beverage in the most extreme circumstances. The *Gelderlandt* had been at sea for five months when the water was first sampled:

2nd August We dranke the last Beere, and we beganne our first allowance to drinke water, four mutskins or measures everie day, and three of wine. [Four mutskins were equal to one pint; a mutskin was about 150 mm or a standard drink of wine.]

What was the diet of a seaman on a long voyage to or from the Indies at the beginning of the 17th century? The Age of Reason and the development of modern Homo inventorensis the compulsive listmaker were just beginning, so archived comprehensive lists of provisions are rare. The most complete list of foodstuffs can be compiled by noting everything that the ships run out of according to their journal records.

The ships *Amsterdam* and *Utrecht* were running short of provisions when they reached the island of Ternate hoping to load a full cargo of cloves in May 1599. Negotiations to barter European goods such as woollen cloth for cloves proved difficult and perhaps too little effort was put into provisioning.

“The 10. day [June] our dyet was shortened, to wit, every day once flesh or fish, and the other three meales Ryce.”

Four meals a day seems copious but only rice and meat or fish would be dull. I'm reminded of the head steward on Terry Southern's uproarious S.S. *Magic Christian* announcing after three days at sea that due to an oversight in the catering department there is nothing left except potatoes.

A few weeks later the journal records that there is nothing left except dried bread, yet between then and 16th November when fresh provisions were finally acquired, they also finished their last supplies of fish, flesh, oil, cheese, honey and smoked meat.

Working back through the list of ration tightening we can see that the normal arrangement was four meals a day with meat or fish (which could be fresh, pickled, or smoked) in at least two meals — in other words, frequent meals, lots of protein, and plenty of drink, much like the majority of passengers on a modern cruise liner.

Pathophysiology of scurvy

Humans share with other primates and with guinea pigs the inability to synthesize ascorbic acid (vitamin C) and therefore require it in their diet. The enzyme which in other species would catalyse the conversion of L-gluconogammalactone to L-ascorbic acid is defective due to a mutation.

Vitamin C is a redox agent, reducing metal ions in many enzymes and removing free radicals, in particular, the Fe² ions required for collagen synthesis change to the more stable Fe³ in the absence of vitamin C. Lack of collagen causes capillary fragility, poor wound healing and other symptoms of scurvy's onset.

Today, according to a medical text, scurvy is not much a problem for mariners but is mostly seen in elderly people “who may be on a tea and toast diet”.